

Synthetic Control

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bookdown::render_book()

Overview

Topics:

1. Synthetic Control
2. Randomized Inference

Chapter 1

Synthetic Control

This is just a generalization of the difference-in-differences design and one of the most important evaluation designs in recent years (Athey and Imben, 2017). We use a quantitative comparative case study by focusing on a particular single unit, such as a state, school, county, group, etc. We optimize a set of weights from a donor pool of comparison states to generate a counterfactual for the treatment state.

The Premise: Abadie and Gardeazabal (2003) use a method of a weighted average of units from a donor pool to model a counterfactual (synthetic control)

1.1 The estimator

Our synthetic control method is the following:

$$\hat{\delta} = Y_{1t} - \sum_{j=2}^{J+1} w_j^* Y_{jt}$$

Where

- $J = 1$ is our treatment group
- $J + 1$ is our donor pool of comparison
- Y_{1t} is the outcome of the treatment group
- Y_{jt} is the outcome from donor j
- w_j^* is a vector of optimized weights that is a function of our choice of covariates

The synthetic control estimator estimates the effect of the program at time $t \geq T_0$.

It's just the difference between our treatment group and a weighted control group at time T_0 . Where our optimized-weighted control group will depend on our set of covariates we choose

Matching variables: Matching variables X_1 and X_0 are chosen as predictors of post-intervention outcomes and must be unaffected by the program, treatment, or intervention.

1.2 *synth* package

First, we will install the *synth* package.

```
ssc install synth
```

Next, we will install the *mat2txt* package.

```
ssc install mat2txt
```

1.3 Prison Construction

We will recreate Cunningham and Kang (2019) work on the effect of building prisons and black male incarceration rates. We will import and inspect our data after installing our packages.

Cunningham (2021) provides the change in prison in Texas and shows that 1993 begins the prison construction boom.

Next, let's bring the data and summarize the black male incarceration rate per 100,000.

```
cd "/Users/Sam/Desktop/Econ 672/Data"
use texas, clear

sum bmprison, detail
```

```
/Users/Sam/Desktop/Econ 672/Data
```

BM Prison				
Percentiles			Smallest	
1%	0	0		
5%	24	0		
10%	47	0	Obs	816

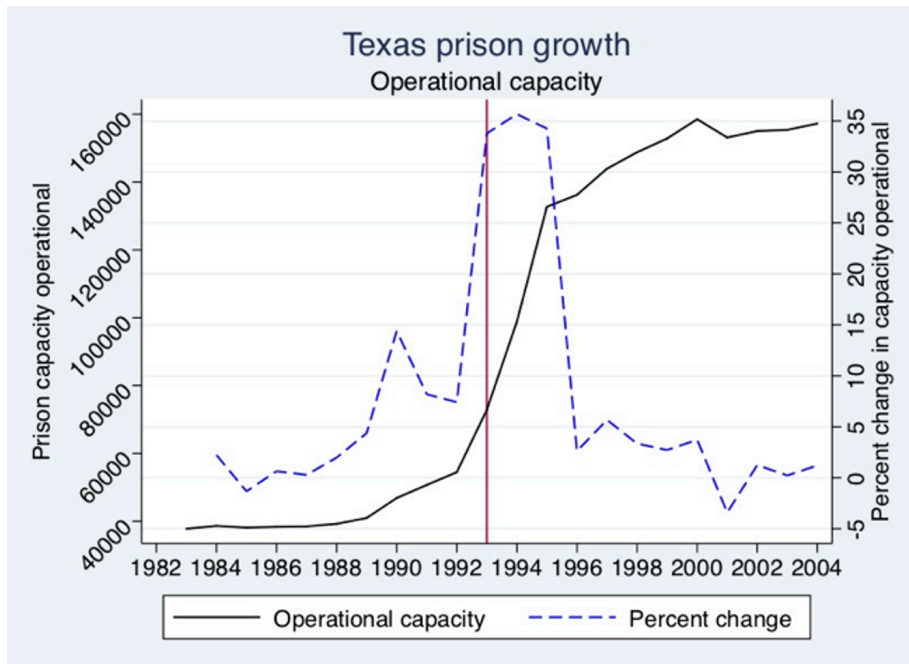


Figure 1.1: Construction of prisons in Texas

25%	489	0	Sum of wgt.	816
50%	3055.5		Mean	7625.753
75%	11425.5	Largest	Std. dev.	10088.11
90%	21153	58393	Variance	1.02e+08
95%	27204	59709	Skewness	2.080135
99%	46235	60785	Kurtosis	8.401005
		61861		

We can see that the mean black male incarceration rate is 3,055 per 100,000.

Next we need to set our idcount and check for redundant values. This is a good practice when working with panel data.

```
sort statefip year
by statefip year: gen idcount = _N
tab idcount

*No redundant or repeated values
drop idcount
```

idcount	Freq.	Percent	Cum.
---------	-------	---------	------

-----+-----			
1	816	100.00	100.00
-----+-----			
Total	816	100.00	

Now, we can set the panel with `statefip` being our unit of analysis and `year` being our time dimension

```
xtset statefip year
```

```
Panel variable: statefip (strongly balanced)
Time variable: year, 1985 to 2000
Delta: 1 unit
```

We have a balanced panel to begin our synthetic control. This step is not necessary for the analysis, but I recommend establishing your panel before further analysis.

1.4 Estimate the Weighted Treatment Effect

We will next calculate the gaps, or the weighted average treatment effect on the treated compared to a synthetic comparison.

We will use our outcome, but also lags in our outcomes, covariates, and lags in our covariates. We will specify that Texas, $FIPS = 48$, is our treatment unit and the time of treatment is 1993. We specify the pre-intervention period from 1985 to 1993 and increment by 1 year with `mseperiod`. This is the period where we want to minimize the Mean Squared Predicted Error or *MSPE*.

Cunningham provides the covariates used to estimate synthetic Texas

- Lagged pre-treatment rates in 1988, 1990, 1991, and 1992
- Unemployment rate, income, and poverty
- Lagged AIDS per capita in 1990, 1991
- Lagged Black population % in 1990, 1991, and 1992
- Percentage of 15-24 year olds in 1990

```
synth bmprison bmprison(1988) bmprison(1990) bmprison(1991) bmprison(1992) alcohol(1988)
aidscapita(1990) aidscapita(1991) income ur poverty black(1990) black(1991) black(1992)
perc1519(1990), trunit(48) trperiod(1993) unitnames(state) mseperiod(1985(1)1993)
resultsperiod(1985(1)2000) keep(synth_bmprate.dta) replace
```

Synthetic Control Method for Comparative Case Studies

First Step: Data Setup

Data Setup successful

Treated Unit: Texas

Control Units: Alabama, Alaska, Arizona, Arkansas, California, Colorado, Connecticut, Delaware, District of Columbia, Florida, Georgia, Hawaii, Idaho, Illinois, Indiana, Iowa, Kansas, Kentucky, Louisiana, Maine, Maryland, Massachusetts, Michigan, Minnesota, Mississippi, Missouri, Montana, Nebraska, Nevada, New Hampshire, New Jersey, New Mexico, New York, North Carolina, North Dakota, Ohio, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, South Dakota, Tennessee, Utah, Vermont, Virginia, Washington, West Virginia, Wisconsin, Wyoming

Dependent Variable: bmprison

MSPE minimized for periods: 1985 1986 1987 1988 1989 1990 1991 1992 1993

Results obtained for periods: 1985 1986 1987 1988 1989 1990 1991 1992 1993 1994
1995 1996 1997 1998 1999 2000

Predictors: bmprison(1988) bmprison(1990) bmprison(1991)
bmprison(1992) alcohol(1990) aidscapita(1990)
aidscapita(1991) income ur poverty black(1990)
black(1991) black(1992) perc1519(1990)

Unless period is specified

predictors are averaged over: 1985 1986 1987 1988 1989 1990 1991 1992

Second Step: Run Optimization

Optimization done

Third Step: Obtain Results

Loss: Root Mean Squared Prediction Error

RMSPE | 1295.489

Unit Weights:

Co_No	Unit_Weight
Alabama	0
Alaska	0
Arizona	0
Arkansas	0
California	.408
Colorado	0
Connecticut	0
Delaware	0
District of Columbia	0
Florida	.109
Georgia	0
Hawaii	0
Idaho	0
Illinois	.36
Indiana	0
Iowa	0
Kansas	0
Kentucky	0
Louisiana	.122
Maine	0
Maryland	0
Massachusetts	0
Michigan	0
Minnesota	0
Mississippi	0
Missouri	0
Montana	0
Nebraska	0
Nevada	0
New Hampshire	0
New Jersey	0
New Mexico	0
New York	0
North Carolina	0
North Dakota	0
Ohio	0
Oklahoma	0
Oregon	0

Pennsylvania		0
Rhode Island		0
South Carolina		0
South Dakota		0
Tennessee		0
Utah		0
Vermont		0
Virginia		0
Washington		0
West Virginia		0
Wisconsin		0
Wyoming		0

Predictor Balance:

		Treated	Synthetic

bmprison(1988)		16956	18374.28
bmprison(1990)		22634	23134.8
bmprison(1991)		23249	24272.24
bmprison(1992)		27568	25776.34
alcohol(1990)		2.47	2.70996
aidsapita(1990)		13.58992	14.99957
aidsapita(1991)		14.83504	17.24535
income		16108.63	18614.54
ur		7.426042	7.188876
poverty		17.2	14.86484
black(1990)		16.15459	15.98038
black(1991)		16.29985	16.16835
black(1992)		16.45955	16.37467
perc1519(1990)		7.703001	7.022702

counter		pri_inf		dual_inf		pri_obj		dual_obj		sigfig		alpha		nu

0		9.51e+01		1.57e-03		-1.82e+00		-5.02e+02		0.000		0.0000		1.00e+02
1		5.46e-01		8.99e-06		-1.77e+00		-9.08e+02		0.000		0.9943		2.73e-05
2		1.14e-02		1.88e-07		-1.41e+00		-3.35e+01		0.000		0.9791		3.67e-05
3		1.21e-03		1.99e-08		-1.45e+00		-4.88e+00		0.000		0.8943		3.09e-05
4		2.39e-04		3.94e-09		-1.77e+00		-2.65e+00		0.497		0.8022		1.15e-05
5		1.14e-05		1.87e-10		-1.81e+00		-1.87e+00		1.673		0.9525		1.67e-07
6		4.95e-06		8.16e-11		-1.81e+00		-1.84e+00		2.042		0.5645		1.01e-06
7		1.77e-06		2.91e-11		-1.82e+00		-1.83e+00		2.490		0.6428		2.88e-07

8		7.32e-07		1.20e-11		-1.82e+00		-1.82e+00		2.865		0.5864		1.39e-07
9		3.15e-07		5.19e-12		-1.82e+00		-1.82e+00		3.238		0.5696		6.39e-08
10		9.21e-09		1.52e-13		-1.82e+00		-1.82e+00		4.622		0.9708		1.16e-10
11		4.60e-11		7.66e-16		-1.82e+00		-1.82e+00		6.907		0.9950		1.39e-13
12		2.30e-13		3.47e-17		-1.82e+00		-1.82e+00		9.204		0.9950		7.23e-16
13		1.31e-15		2.05e-17		-1.82e+00		-1.82e+00		11.502		0.9950		3.64e-18
14		6.19e-16		3.65e-17		-1.82e+00		-1.82e+00		13.792		0.9950		1.84e-20

optimization converged

We have a few interesting results.

First, our $RMSPE = 1295.489$. We can and should continue to test the combination of covariates that reduces the $RMSPE$ in the pre-treatment period. Recall that Abadie, Diamond, and Hainmueller suggest choosing a V that minimizes the mean squared prediction error in the pre-treatment period. Recall that we can influence the w through V since V is a function of our covariates.

$$RMSPE = \sqrt{\sum_{t=1}^{T_0} \left(Y_{1t} - \sum_{j=2}^{J+1} w_j^*(V) Y_{jt} \right)^2}$$

Second, our results show that California is the largest donor with a weight of 40.8%. Illinois is next with a weight of 36%, while Louisiana and Florida are the third and fourth largest donors at 12.2% and 10.9%, respectively.

Next we have our predictor balance between Texas and Synthetic Texas.

These looked balanced, but what are we missing? Ways to test the null hypothesis that these are not different!

1.5 Plot the Graph

We can add the option *figure* to the *synth* command to plot the Texas vs Synthetic Texas.

```
synth    bmprison bmprison(1988) bmprison(1990) bmprison(1991) bmprison(1992) alcohol(1992)
```

1.6 Plot the Gap or Difference

We are going to pull our results from the synthetic control method with the file *synth_bmprate.dta*, which we defined in the *keep* option.

Predictor Balance:

	Treated	Synthetic
bmprison(1988)	16956	18374.28
bmprison(1990)	22634	23134.8
bmprison(1991)	23249	24272.24
bmprison(1992)	27568	25776.34
alcohol(1990)	2.47	2.70996
aids capita(1990)	13.58992	14.99957
aids capita(1991)	14.83504	17.24535
income	16108.63	18614.54
ur	7.426042	7.188876
poverty	17.2	14.86484
black(1990)	16.15459	15.98038
black(1991)	16.29985	16.16835
black(1992)	16.45955	16.37467
perc1519(1990)	7.703001	7.022702

Figure 1.2: Balance between Texas and Synthetic Texas

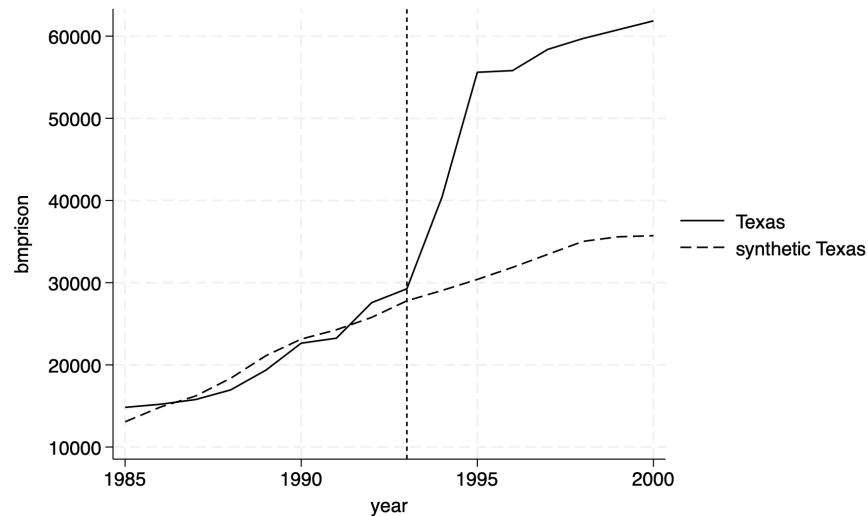


Figure 1.3: Synthetic Texas vs Texas

```
use synth_bmprate.dta, clear
```

The first two columns are our donor pool and their weights. We don't need these now so we can drop them. The 3rd and 4th columns are our outcomes of interest: Y of Texas and Y of Synthetic Texas. The 5 column is our time period

```
keep _Y_treated _Y_synthetic _time
drop if _time==.
rename _time year
rename _Y_treated treat
rename _Y_synthetic counterfact
```

Generate the difference/gap between Texas outcome and Synthetic Texas outcome.

```
gen gap48=treat-counterfact
sort year
```

Finally, we will plot the difference between two outcomes

```
twoway (line gap48 year,lp(solid)lw(vthin)lcolor(black)), yline(0, lpattern(shortdash)
xline(1993, lpattern(shortdash) lcolor(black)) xtitle("Year",si(medsmall)) xlabel(
yttitle("Gap in black male prisoners", size(medsmall)) legend(off)
```



```
save synth_bmprate_48.dta, replace
```

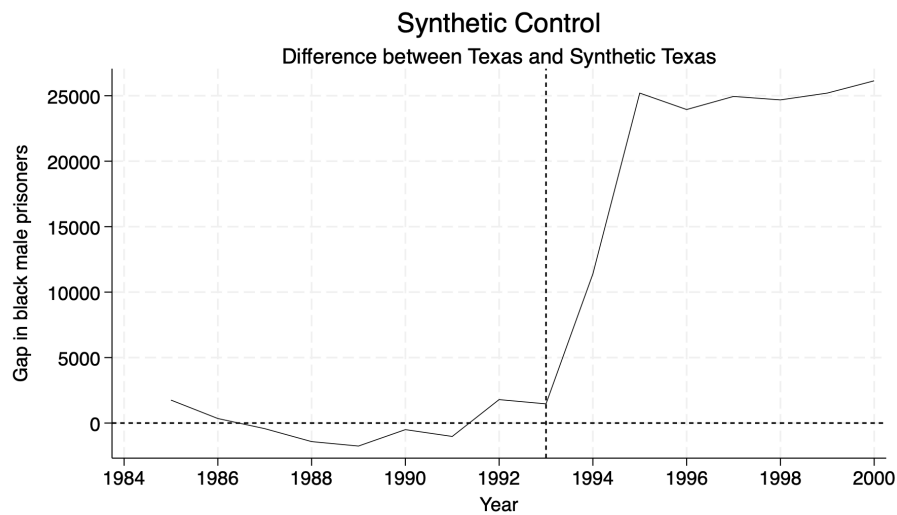


Figure 1.4: Difference between Texas and Synthetic Texas

We can see an difference about 25,000 black males incarcerated per capita in 1995. However, we need to test the null hypothesis to see if this is a statistically significant effect.

Chapter 2

Randomized Inference

We have estimated the difference between Texas and Synthetic Texas, but we have not been able to infer the impact. We will use randomized inference to test our hypothesis.

We will establish a null hypothesis of “no treatment effect”, and then use randomized inference to test the hypothesis.

Our process to get exact p -values will be the following:

1. Loop through each state to calculate placebo treatments
2. Calculate the $RMSPE$ for each placebo state in the pre-treatment period
3. Calculate the $RMSPE$ for each placebo state in the post-treatment period
4. Compute the ratio of $\frac{post-treatment}{pre-treatment}$
5. Sort this ratio in descending order
6. Calculate the treatment units ratio in the distribution of $p = \frac{RANK}{Total}$

2.1 Placebo Tests

First, we will set local macro with fips codes for the 39 states used. We will want to include Texas as part of the donor pool for each placebo test.

```
local statelist 1 2 4 5 6 8 9 10 11 12 13 15 16 17 18 20 21 22 23 24 25 26 27 28 ///
                29 30 31 32 33 34 35 36 37 38 39 40 41 42 45 46 47 48 49 51 53 55
```

2.1.1 Caculate $RMSPE$ for each state

Next, we will calculate the Root Mean Squared Prediction Error $RMSPE$ for each state in the donor pool for each time period.

```

foreach i of local statelist {
synth  bmprison ///
        bmprison(1990) bmprison(1992) bmprison(1991) bmprison(1988) ///
        alcohol(1990) aidscapita(1990) aidscapita(1991) ///
        income ur poverty black(1990) black(1991) black(1992) ///
        perc1519(1990), ///
        trunit(`i') trperiod(1993) unitnames(state) ///
        mspeperiod(1985(1)1993) resultsperiod(1985(1)2000) ///
        keep(synth_bmprate_`i'.dta) replace
/* check the V matrix */
matrix state`i' = e(RMSPE)
}

```

```

foreach i of local statelist {
matrix rownames state`i'=`i'
quietly matlist state`i', names(rows)
}

```

counter	pri_inf	dual_inf	pri_obj	dual_obj	sigfig	alpha	nu
0	9.51e+01	9.77e-04	-1.61e+00	-5.02e+02	0.000	0.0000	1.00e+02
1	4.76e-01	4.88e-06	-1.59e+00	-9.08e+02	0.000	0.9950	2.07e-05
2	6.46e-03	6.63e-08	-1.41e+00	-2.55e+01	0.000	0.9864	1.52e-05
3	1.08e-03	1.11e-08	-1.47e+00	-5.50e+00	0.000	0.8333	5.78e-05
4	4.51e-05	4.63e-10	-1.58e+00	-1.80e+00	1.070	0.9581	5.91e-07
5	5.36e-06	5.51e-11	-1.62e+00	-1.67e+00	1.659	0.8812	2.62e-07
6	8.17e-07	8.39e-12	-1.62e+00	-1.63e+00	2.468	0.8476	1.13e-07
7	6.15e-07	6.32e-12	-1.62e+00	-1.62e+00	2.590	0.2470	4.81e-07
8	7.65e-08	7.85e-13	-1.62e+00	-1.62e+00	3.434	0.8757	8.79e-09
9	4.58e-08	4.70e-13	-1.62e+00	-1.62e+00	3.597	0.4009	3.20e-08
10	3.69e-09	3.79e-14	-1.62e+00	-1.62e+00	4.510	0.9194	3.61e-10
11	2.55e-09	2.62e-14	-1.62e+00	-1.62e+00	4.684	0.3096	3.63e-09
12	9.77e-10	1.00e-14	-1.62e+00	-1.62e+00	5.105	0.6169	7.06e-10
13	6.62e-12	7.59e-17	-1.62e+00	-1.62e+00	6.679	0.9932	7.80e-14
14	3.42e-14	2.46e-17	-1.62e+00	-1.62e+00	8.961	0.9948	1.21e-15
15	7.23e-16	1.56e-17	-1.62e+00	-1.62e+00	11.259	0.9950	5.92e-18
16	6.73e-16	2.00e-17	-1.62e+00	-1.62e+00	13.538	0.9950	2.98e-20

optimization converged

counter	pri_inf	dual_inf	pri_obj	dual_obj	sigfig	alpha	nu
0	9.51e+01	2.04e-02	-5.74e-02	-5.00e+02	0.000	0.0000	1.00e+02
1	6.07e-01	1.30e-04	-5.41e-02	-9.06e+02	0.000	0.9936	3.37e-05
2	3.04e-03	6.52e-07	4.93e-02	-1.87e+01	0.000	0.9950	2.11e-06
3	8.62e-05	1.85e-08	3.36e-02	-5.02e-01	0.285	0.9716	1.27e-06

4		3.13e-06		6.72e-10		-4.33e-02		-9.81e-02		1.279		0.9637		5.89e-08
5		1.60e-06		3.43e-10		-4.98e-02		-7.66e-02		1.593		0.4900		1.30e-06
6		5.13e-07		1.10e-10		-5.41e-02		-6.28e-02		2.085		0.6785		2.43e-07
7		1.05e-07		2.25e-11		-5.65e-02		-5.87e-02		2.688		0.7959		3.10e-08
8		4.15e-08		8.91e-12		-5.72e-02		-5.81e-02		3.098		0.6040		3.02e-08
9		9.42e-09		2.02e-12		-5.76e-02		-5.78e-02		3.697		0.7730		3.75e-09
10		1.48e-10		3.18e-14		-5.77e-02		-5.77e-02		4.850		0.9843		4.35e-12
11		8.62e-13		2.05e-16		-5.77e-02		-5.77e-02		6.874		0.9942		4.19e-14
12		4.32e-15		8.73e-17		-5.77e-02		-5.77e-02		9.173		0.9950		2.92e-16
13		7.54e-16		1.20e-16		-5.77e-02		-5.77e-02		11.471		0.9950		1.47e-18
14		7.61e-16		1.29e-16		-5.77e-02		-5.77e-02		13.765		0.9950		7.40e-21

optimization converged

counter		pri_inf		dual_inf		pri_obj		dual_obj		sigfig		alpha		nu

0		9.51e+01		1.85e-02		-5.42e-02		-5.00e+02		0.000		0.0000		1.00e+02
1		6.54e-01		1.27e-04		-5.16e-02		-9.06e+02		0.000		0.9931		3.91e-05
2		3.27e-03		6.36e-07		4.11e-02		-1.95e+01		0.000		0.9950		2.13e-06
3		9.06e-05		1.76e-08		2.64e-02		-5.19e-01		0.274		0.9723		1.26e-06
4		4.19e-06		8.15e-10		-4.64e-02		-1.06e-01		1.243		0.9538		9.71e-08
5		3.11e-07		6.06e-11		-5.40e-02		-6.08e-02		2.186		0.9256		2.77e-08
6		2.73e-07		5.31e-11		-5.43e-02		-6.04e-02		2.244		0.1242		5.14e-07
7		1.99e-07		3.87e-11		-5.46e-02		-5.90e-02		2.386		0.2704		3.04e-07
8		1.66e-08		3.22e-12		-5.51e-02		-5.56e-02		3.266		0.9167		2.52e-09
9		1.70e-09		3.31e-13		-5.51e-02		-5.51e-02		4.216		0.8973		5.07e-10
10		9.26e-10		1.80e-13		-5.51e-02		-5.51e-02		4.463		0.4557		1.74e-09
11		3.19e-10		6.22e-14		-5.51e-02		-5.51e-02		4.878		0.6550		3.81e-10
12		1.43e-10		2.78e-14		-5.51e-02		-5.51e-02		5.225		0.5525		2.51e-10
13		4.33e-11		8.42e-15		-5.51e-02		-5.51e-02		5.732		0.6972		5.03e-11
14		5.86e-13		1.72e-16		-5.51e-02		-5.51e-02		7.034		0.9865		2.97e-14
15		4.27e-14		1.12e-16		-5.51e-02		-5.51e-02		8.175		0.9272		4.34e-14
16		7.36e-16		9.42e-17		-5.51e-02		-5.51e-02		10.101		0.9909		4.84e-17
17		7.84e-16		7.92e-17		-5.51e-02		-5.51e-02		12.034		0.9945		2.10e-19

optimization converged

counter		pri_inf		dual_inf		pri_obj		dual_obj		sigfig		alpha		nu

0		9.51e+01		1.98e-03		-5.97e-01		-5.01e+02		0.000		0.0000		1.00e+02
1		5.74e-01		1.19e-05		-5.90e-01		-9.07e+02		0.000		0.9940		3.01e-05
2		4.44e-03		9.24e-08		-5.55e-01		-2.11e+01		0.000		0.9923		5.02e-06
3		2.85e-04		5.94e-09		-5.60e-01		-1.89e+00		0.069		0.9357		7.22e-06
4		3.41e-05		7.09e-10		-5.82e-01		-7.48e-01		0.980		0.8806		1.60e-06
5		1.11e-05		2.30e-10		-5.97e-01		-6.58e-01		1.415		0.6751		1.53e-06
6		8.46e-07		1.76e-11		-5.99e-01		-6.05e-01		2.451		0.9235		3.01e-08
7		8.05e-08		1.67e-12		-5.99e-01		-6.00e-01		3.341		0.9049		4.30e-09
8		5.50e-08		1.15e-12		-6.00e-01		-6.00e-01		3.431		0.3162		3.20e-08

9	4.43e-09	9.23e-14	-6.00e-01	-6.00e-01	4.396	0.9194	3.23e-10
10	2.21e-09	4.59e-14	-6.00e-01	-6.00e-01	4.715	0.5023	1.44e-09
11	2.13e-09	4.44e-14	-6.00e-01	-6.00e-01	4.727	0.0337	2.86e-09
12	4.64e-10	9.65e-15	-6.00e-01	-6.00e-01	5.367	0.7826	1.22e-10
13	1.29e-11	2.66e-16	-6.00e-01	-6.00e-01	6.813	0.9722	4.42e-13
14	6.45e-14	2.79e-17	-6.00e-01	-6.00e-01	9.098	0.9950	5.08e-16
15	8.24e-16	3.53e-17	-6.00e-01	-6.00e-01	11.396	0.9950	2.64e-18
16	6.52e-16	2.49e-17	-6.00e-01	-6.00e-01	13.704	0.9950	1.33e-20

optimization converged							
counter	pri_inf	dual_inf	pri_obj	dual_obj	sigfig	alpha	nu

0	9.51e+01	2.58e-03	-2.52e+00	-5.02e+02	0.000	0.0000	1.00e+02
1	1.23e+00	3.35e-05	-2.37e+00	-9.05e+02	0.000	0.9870	1.40e-04
2	4.81e-02	1.30e-06	-1.45e+00	-6.07e+01	0.000	0.9611	1.44e-04
3	3.83e-03	1.04e-07	-1.47e+00	-6.75e+00	0.000	0.9204	3.37e-05
4	1.69e-03	4.58e-08	-2.10e+00	-4.63e+00	0.089	0.5594	9.27e-05
5	1.07e-04	2.90e-09	-2.30e+00	-2.53e+00	1.167	0.9367	8.52e-07
6	2.34e-05	6.36e-10	-2.36e+00	-2.41e+00	1.796	0.7807	9.32e-07
7	2.27e-05	6.17e-10	-2.36e+00	-2.41e+00	1.809	0.0303	5.02e-06
8	8.66e-06	2.35e-10	-2.37e+00	-2.39e+00	2.241	0.6190	6.71e-07
9	1.02e-07	2.75e-12	-2.38e+00	-2.38e+00	3.937	0.9883	2.21e-10
10	5.12e-10	1.39e-14	-2.38e+00	-2.38e+00	6.230	0.9950	8.24e-13
11	2.56e-12	6.50e-17	-2.38e+00	-2.38e+00	8.527	0.9950	4.12e-15
12	1.28e-14	2.00e-17	-2.38e+00	-2.38e+00	10.825	0.9950	2.08e-17
13	6.22e-16	2.12e-17	-2.38e+00	-2.38e+00	13.124	0.9950	1.05e-19

optimization converged							
counter	pri_inf	dual_inf	pri_obj	dual_obj	sigfig	alpha	nu

0	9.51e+01	1.64e-02	-6.51e-02	-5.00e+02	0.000	0.0000	1.00e+02
1	5.57e-01	9.61e-05	-6.10e-02	-9.06e+02	0.000	0.9941	2.84e-05
2	2.78e-03	4.81e-07	2.02e-02	-1.78e+01	0.000	0.9950	2.09e-06
3	7.62e-05	1.32e-08	6.32e-03	-4.86e-01	0.311	0.9726	1.12e-06
4	3.95e-06	6.82e-10	-5.70e-02	-1.12e-01	1.285	0.9482	1.10e-07
5	1.66e-06	2.86e-10	-6.24e-02	-8.47e-02	1.679	0.5808	8.61e-07
6	7.33e-07	1.27e-10	-6.44e-02	-7.41e-02	2.043	0.5573	3.91e-07
7	3.72e-07	6.42e-11	-6.58e-02	-7.07e-02	2.344	0.4929	2.25e-07
8	7.36e-08	1.27e-11	-6.65e-02	-6.76e-02	2.995	0.8021	1.62e-08
9	2.14e-08	3.70e-12	-6.67e-02	-6.70e-02	3.515	0.7091	7.96e-09
10	5.71e-09	9.86e-13	-6.68e-02	-6.69e-02	4.003	0.7332	2.01e-09
11	2.94e-09	5.08e-13	-6.68e-02	-6.69e-02	4.302	0.4845	2.56e-09
12	1.20e-10	2.07e-14	-6.68e-02	-6.68e-02	5.067	0.9593	7.33e-12
13	2.58e-12	4.51e-16	-6.68e-02	-6.68e-02	6.574	0.9784	3.52e-13
14	1.29e-14	6.22e-17	-6.68e-02	-6.68e-02	8.842	0.9950	5.95e-16
15	9.26e-16	6.45e-17	-6.68e-02	-6.68e-02	11.140	0.9950	3.17e-18

16 | 7.70e-16 | 7.65e-17 | -6.68e-02 | -6.68e-02 | 13.436 | 0.9950 | 1.60e-20

optimization converged

counter	pri_inf	dual_inf	pri_obj	dual_obj	sigfig	alpha	nu
0	9.51e+01	3.96e-03	-2.30e-01	-5.00e+02	0.000	0.0000	1.00e+02
1	4.76e-01	1.98e-05	-2.29e-01	-9.07e+02	0.000	0.9950	2.07e-05
2	2.38e-03	9.89e-08	-2.17e-01	-1.66e+01	0.000	0.9950	2.06e-06
3	3.56e-05	1.48e-09	-2.20e-01	-4.66e-01	0.694	0.9850	3.06e-07
4	1.74e-06	7.24e-11	-2.29e-01	-2.45e-01	1.879	0.9511	4.92e-08
5	1.40e-07	5.81e-12	-2.30e-01	-2.32e-01	2.775	0.9198	8.77e-09
6	4.64e-08	1.93e-12	-2.30e-01	-2.31e-01	3.247	0.6682	1.99e-08
7	1.87e-08	7.78e-13	-2.30e-01	-2.31e-01	3.649	0.5964	1.01e-08
8	6.16e-09	2.56e-13	-2.30e-01	-2.30e-01	4.117	0.6709	2.63e-09
9	4.89e-10	2.04e-14	-2.30e-01	-2.30e-01	5.014	0.9206	4.97e-11
10	1.67e-11	6.87e-16	-2.30e-01	-2.30e-01	6.418	0.9658	1.16e-12
11	8.37e-14	6.01e-17	-2.30e-01	-2.30e-01	8.698	0.9950	9.71e-16
12	8.04e-16	4.97e-17	-2.30e-01	-2.30e-01	10.996	0.9950	5.10e-18
13	7.37e-16	4.75e-17	-2.30e-01	-2.30e-01	13.293	0.9950	2.57e-20

optimization converged

counter	pri_inf	dual_inf	pri_obj	dual_obj	sigfig	alpha	nu
0	9.51e+01	1.98e-03	-6.93e-01	-5.01e+02	0.000	0.0000	1.00e+02
1	7.06e-01	1.47e-05	-6.85e-01	-9.06e+02	0.000	0.9926	4.56e-05
2	8.38e-03	1.74e-07	-6.11e-01	-2.72e+01	0.000	0.9881	1.21e-05
3	7.09e-04	1.47e-08	-6.20e-01	-2.89e+00	0.000	0.9154	1.63e-05
4	3.06e-05	6.36e-10	-6.67e-01	-7.88e-01	1.139	0.9568	3.53e-07
5	4.84e-06	1.01e-10	-6.87e-01	-7.12e-01	1.832	0.8420	2.57e-07
6	4.13e-07	8.59e-12	-6.93e-01	-6.97e-01	2.650	0.9146	1.52e-08
7	7.21e-09	1.50e-13	-6.94e-01	-6.95e-01	3.566	0.9826	9.55e-11
8	1.20e-09	2.50e-14	-6.94e-01	-6.94e-01	4.239	0.8331	1.09e-09
9	7.75e-10	1.61e-14	-6.94e-01	-6.94e-01	4.452	0.3557	3.78e-09
10	5.45e-10	1.13e-14	-6.94e-01	-6.94e-01	4.621	0.2975	2.79e-09
11	1.57e-10	3.26e-15	-6.94e-01	-6.94e-01	4.852	0.7114	2.95e-10
12	2.78e-12	4.98e-17	-6.94e-01	-6.94e-01	5.966	0.9823	6.18e-13
13	3.13e-14	3.78e-17	-6.94e-01	-6.94e-01	7.192	0.9887	1.93e-14
14	7.55e-16	3.34e-17	-6.94e-01	-6.94e-01	9.361	0.9949	2.31e-16
15	5.18e-16	3.59e-17	-6.94e-01	-6.94e-01	11.660	0.9950	1.52e-18
16	8.32e-16	4.21e-17	-6.94e-01	-6.94e-01	13.977	0.9950	7.67e-21

optimization converged

counter	pri_inf	dual_inf	pri_obj	dual_obj	sigfig	alpha	nu
0	9.51e+01	9.41e-04	-1.12e+01	-5.10e+02	0.000	0.0000	1.00e+02
1	1.58e+00	1.56e-05	-1.06e+01	-9.10e+02	0.000	0.9834	2.28e-04

2	1.42e-01	1.40e-06	-6.25e+00	-1.14e+02	0.000	0.9103	8.10e-04
3	7.03e-03	6.96e-08	-5.68e+00	-1.36e+01	0.000	0.9503	2.51e-05
4	1.89e-03	1.87e-08	-7.77e+00	-1.01e+01	0.569	0.7310	5.14e-05
5	3.64e-05	3.60e-10	-8.34e+00	-8.41e+00	2.122	0.9808	7.41e-08
6	1.82e-07	1.80e-12	-8.38e+00	-8.38e+00	4.407	0.9950	1.47e-10
7	9.09e-10	9.00e-15	-8.38e+00	-8.38e+00	6.705	0.9950	7.67e-13
8	4.55e-12	4.61e-17	-8.38e+00	-8.38e+00	9.002	0.9950	3.87e-15
9	2.28e-14	1.34e-17	-8.38e+00	-8.38e+00	11.300	0.9950	1.95e-17
10	5.76e-16	1.10e-17	-8.38e+00	-8.38e+00	13.596	0.9950	9.83e-20

optimization converged

counter	pri_inf	dual_inf	pri_obj	dual_obj	sigfig	alpha	nu

0	9.51e+01	1.66e-03	-1.84e+00	-5.02e+02	0.000	0.0000	1.00e+02
1	5.43e-01	9.46e-06	-1.79e+00	-9.08e+02	0.000	0.9943	2.70e-05
2	1.20e-02	2.10e-07	-1.41e+00	-3.45e+01	0.000	0.9778	4.12e-05
3	1.32e-03	2.30e-08	-1.46e+00	-5.14e+00	0.000	0.8901	3.45e-05
4	2.27e-04	3.95e-09	-1.80e+00	-2.66e+00	0.513	0.8285	9.26e-06
5	1.09e-05	1.90e-10	-1.84e+00	-1.90e+00	1.681	0.9519	1.66e-07
6	4.31e-06	7.50e-11	-1.84e+00	-1.87e+00	2.097	0.6052	8.21e-07
7	1.80e-06	3.14e-11	-1.85e+00	-1.85e+00	2.483	0.5813	3.57e-07
8	1.32e-07	2.30e-12	-1.85e+00	-1.85e+00	3.231	0.9267	4.21e-09
9	4.33e-08	7.54e-13	-1.85e+00	-1.85e+00	3.718	0.6724	1.58e-08
10	2.58e-08	4.49e-13	-1.85e+00	-1.85e+00	3.924	0.4045	1.79e-08
11	1.33e-09	2.32e-14	-1.85e+00	-1.85e+00	5.148	0.9483	7.55e-11
12	7.12e-12	1.17e-16	-1.85e+00	-1.85e+00	7.381	0.9947	4.79e-14
13	3.56e-14	2.13e-17	-1.85e+00	-1.85e+00	9.679	0.9950	2.45e-16
14	8.10e-16	2.06e-17	-1.85e+00	-1.85e+00	11.977	0.9950	1.23e-18
15	5.53e-16	1.85e-17	-1.85e+00	-1.85e+00	14.352	0.9950	6.22e-21

optimization converged

counter	pri_inf	dual_inf	pri_obj	dual_obj	sigfig	alpha	nu

0	9.51e+01	7.66e-04	-2.15e+00	-5.02e+02	0.000	0.0000	1.00e+02
1	5.24e-01	4.22e-06	-2.15e+00	-9.08e+02	0.000	0.9945	2.51e-05
2	7.92e-03	6.37e-08	-1.82e+00	-2.81e+01	0.000	0.9849	1.91e-05
3	1.50e-03	1.21e-08	-1.92e+00	-6.93e+00	0.000	0.8104	8.24e-05
4	1.98e-04	1.59e-09	-2.09e+00	-2.80e+00	0.639	0.8683	7.40e-06
5	2.55e-05	2.05e-10	-2.14e+00	-2.25e+00	1.430	0.8711	9.99e-07
6	1.94e-05	1.56e-10	-2.15e+00	-2.23e+00	1.554	0.2383	6.46e-06
7	3.06e-07	2.46e-12	-2.16e+00	-2.17e+00	2.545	0.9842	1.81e-09
8	2.58e-08	2.08e-13	-2.16e+00	-2.16e+00	3.523	0.9157	5.37e-09
9	1.24e-08	1.00e-13	-2.16e+00	-2.16e+00	3.840	0.5177	1.99e-08
10	3.52e-09	2.83e-14	-2.16e+00	-2.16e+00	4.357	0.7174	3.17e-09
11	4.70e-11	3.88e-16	-2.16e+00	-2.16e+00	5.323	0.9866	2.06e-12
12	1.97e-11	1.62e-16	-2.16e+00	-2.16e+00	5.589	0.5809	2.35e-10

13		1.91e-11		1.46e-16		-2.16e+00		-2.16e+00		5.605		0.0326		7.56e-10
14		2.49e-12		2.62e-17		-2.16e+00		-2.16e+00		6.302		0.8693		1.14e-11
15		1.75e-13		2.44e-17		-2.16e+00		-2.16e+00		7.413		0.9296		6.53e-13
16		1.11e-15		1.91e-17		-2.16e+00		-2.16e+00		9.652		0.9948		2.71e-16
17		8.05e-16		1.69e-17		-2.16e+00		-2.16e+00		11.949		0.9950		1.46e-18
18		6.99e-16		1.40e-17		-2.16e+00		-2.16e+00		14.361		0.9950		7.33e-21

optimization converged

counter		pri_inf		dual_inf		pri_obj		dual_obj		sigfig		alpha		nu

0		9.51e+01		1.03e-02		-1.26e-01		-5.00e+02		0.000		0.0000		1.00e+02
1		6.48e-01		7.01e-05		-1.20e-01		-9.06e+02		0.000		0.9932		3.84e-05
2		3.24e-03		3.50e-07		-4.51e-02		-1.95e+01		0.000		0.9950		2.13e-06
3		7.53e-05		8.14e-09		-5.52e-02		-5.11e-01		0.364		0.9768		8.81e-07
4		4.16e-06		4.49e-10		-1.07e-01		-1.52e-01		1.388		0.9448		1.16e-07
5		8.62e-08		9.32e-12		-1.23e-01		-1.30e-01		2.222		0.9793		1.62e-09
6		1.96e-08		2.11e-12		-1.25e-01		-1.27e-01		2.820		0.7731		2.99e-08
7		1.79e-09		1.93e-13		-1.26e-01		-1.27e-01		3.719		0.9085		1.20e-09
8		2.40e-11		2.63e-15		-1.27e-01		-1.27e-01		5.416		0.9866		3.20e-12
9		1.20e-13		1.32e-16		-1.27e-01		-1.27e-01		7.705		0.9950		9.00e-15
10		9.98e-16		1.25e-16		-1.27e-01		-1.27e-01		10.003		0.9950		4.59e-17
11		7.55e-16		9.69e-17		-1.27e-01		-1.27e-01		12.300		0.9950		2.32e-19

optimization converged

counter		pri_inf		dual_inf		pri_obj		dual_obj		sigfig		alpha		nu

0		9.51e+01		1.80e-01		-3.30e-03		-5.00e+02		0.000		0.0000		1.00e+02
1		8.79e-01		1.67e-03		2.87e-04		-9.05e+02		0.000		0.9908		7.07e-05
2		4.40e-03		8.33e-06		2.38e-01		-2.31e+01		0.000		0.9950		2.22e-06
3		1.58e-04		3.00e-07		2.08e-01		-6.38e-01		0.154		0.9640		2.54e-06
4		1.91e-06		3.62e-09		3.32e-02		-5.12e-02		1.088		0.9879		1.02e-08
5		2.04e-07		3.87e-10		3.87e-03		-1.22e-02		1.795		0.8931		8.13e-08
6		4.05e-09		7.68e-12		-2.59e-03		-4.97e-03		2.624		0.9801		5.26e-10
7		1.01e-09		1.90e-12		-3.39e-03		-4.06e-03		3.178		0.7521		1.27e-08
8		3.11e-10		5.88e-13		-3.61e-03		-3.82e-03		3.671		0.6910		5.56e-09
9		1.26e-10		2.38e-13		-3.67e-03		-3.76e-03		4.077		0.5956		3.12e-09
10		2.35e-11		4.46e-14		-3.72e-03		-3.74e-03		4.701		0.8127		2.52e-10
11		9.90e-12		1.88e-14		-3.72e-03		-3.74e-03		4.724		0.5789		3.17e-10
12		7.05e-13		1.33e-15		-3.73e-03		-3.73e-03		5.566		0.9288		8.04e-12
13		2.91e-14		9.70e-17		-3.73e-03		-3.73e-03		6.667		0.9588		3.85e-13
14		7.05e-16		9.90e-17		-3.73e-03		-3.73e-03		8.429		0.9937		7.20e-16
15		7.93e-16		1.25e-16		-3.73e-03		-3.73e-03		10.728		0.9950		7.73e-18
16		6.30e-16		6.77e-17		-3.73e-03		-3.73e-03		13.026		0.9950		3.88e-20

optimization converged

counter		pri_inf		dual_inf		pri_obj		dual_obj		sigfig		alpha		nu
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0		9.51e+01		1.35e-03		-1.23e+00		-5.01e+02		0.000		0.0000		1.00e+02
1		4.76e-01		6.74e-06		-1.21e+00		-9.08e+02		0.000		0.9950		2.07e-05
2		5.26e-03		7.46e-08		-1.07e+00		-2.29e+01		0.000		0.9889		1.01e-05
3		3.73e-04		5.29e-09		-1.09e+00		-2.65e+00		0.128		0.9291		9.32e-06
4		5.46e-05		7.73e-10		-1.22e+00		-1.54e+00		0.840		0.8539		2.83e-06
5		1.20e-05		1.71e-10		-1.22e+00		-1.29e+00		1.494		0.7794		1.34e-06
6		8.86e-06		1.26e-10		-1.23e+00		-1.28e+00		1.633		0.2637		3.67e-06
7		6.82e-06		9.66e-11		-1.23e+00		-1.27e+00		1.751		0.2309		2.93e-06
8		2.97e-06		4.21e-11		-1.23e+00		-1.25e+00		2.110		0.5644		6.72e-07
9		7.37e-07		1.04e-11		-1.23e+00		-1.23e+00		2.692		0.7519		9.21e-08
10		5.02e-07		7.11e-12		-1.23e+00		-1.23e+00		2.868		0.3190		1.97e-07
11		7.01e-08		9.93e-13		-1.23e+00		-1.23e+00		3.601		0.8603		5.00e-09
12		2.35e-08		3.33e-13		-1.23e+00		-1.23e+00		4.080		0.6644		5.54e-09
13		6.34e-09		8.98e-14		-1.23e+00		-1.23e+00		4.571		0.7305		1.17e-09
14		2.54e-09		3.60e-14		-1.23e+00		-1.23e+00		4.976		0.5991		8.58e-10
15		9.42e-10		1.33e-14		-1.23e+00		-1.23e+00		5.410		0.6294		2.87e-10
16		8.31e-12		1.17e-16		-1.23e+00		-1.23e+00		7.101		0.9912		5.60e-14
17		4.82e-14		1.65e-17		-1.23e+00		-1.23e+00		9.334		0.9942		4.92e-16
18		8.82e-16		1.70e-17		-1.23e+00		-1.23e+00		11.631		0.9950		2.14e-18
19		7.46e-16		1.81e-17		-1.23e+00		-1.23e+00		13.919		0.9950		1.08e-20

optimization converged

counter		pri_inf		dual_inf		pri_obj		dual_obj		sigfig		alpha		nu
0		9.51e+01		4.76e-03		-1.90e-01		-5.00e+02		0.000		0.0000		1.00e+02
1		4.98e-01		2.49e-05		-1.89e-01		-9.07e+02		0.000		0.9948		2.27e-05
2		2.49e-03		1.25e-07		-1.70e-01		-1.69e+01		0.000		0.9950		2.07e-06
3		4.32e-05		2.16e-09		-1.74e-01		-4.67e-01		0.603		0.9827		4.21e-07
4		2.90e-06		1.45e-10		-1.89e-01		-2.15e-01		1.654		0.9328		1.11e-07
5		2.03e-07		1.02e-11		-1.89e-01		-1.92e-01		2.762		0.9299		1.08e-08
6		2.02e-08		1.01e-12		-1.90e-01		-1.90e-01		3.593		0.9008		1.71e-09
7		6.91e-09		3.46e-13		-1.90e-01		-1.90e-01		4.065		0.6574		3.14e-09
8		1.26e-09		6.31e-14		-1.90e-01		-1.90e-01		4.729		0.8176		2.91e-10
9		2.28e-10		1.14e-14		-1.90e-01		-1.90e-01		5.259		0.8189		6.22e-11
10		9.46e-11		4.74e-15		-1.90e-01		-1.90e-01		5.637		0.5856		1.00e-10
11		4.38e-12		2.28e-16		-1.90e-01		-1.90e-01		6.792		0.9537		4.90e-13
12		8.43e-14		4.32e-17		-1.90e-01		-1.90e-01		8.360		0.9808		5.90e-15
13		3.37e-14		4.41e-17		-1.90e-01		-1.90e-01		8.764		0.5995		7.42e-14
14		1.05e-15		5.00e-17		-1.90e-01		-1.90e-01		10.383		0.9827		5.11e-17
15		9.56e-16		4.16e-17		-1.90e-01		-1.90e-01		12.617		0.9943		1.32e-19

optimization converged

counter		pri_inf		dual_inf		pri_obj		dual_obj		sigfig		alpha		nu
0		9.51e+01		1.06e-02		-9.22e-02		-5.00e+02		0.000		0.0000		1.00e+02

1	5.31e-01	5.90e-05	-9.05e-02	-9.06e+02	0.000	0.9944	2.58e-05
2	2.65e-03	2.95e-07	-3.10e-02	-1.74e+01	0.000	0.9950	2.08e-06
3	6.57e-05	7.31e-09	-4.14e-02	-4.75e-01	0.381	0.9752	8.93e-07
4	3.31e-06	3.68e-10	-8.76e-02	-1.31e-01	1.394	0.9497	9.16e-08
5	4.97e-07	5.53e-11	-9.16e-02	-9.88e-02	2.183	0.8498	8.41e-08
6	4.99e-08	5.55e-12	-9.24e-02	-9.35e-02	3.001	0.8995	6.08e-09
7	1.64e-08	1.83e-12	-9.24e-02	-9.27e-02	3.487	0.6711	1.03e-08
8	6.86e-09	7.63e-13	-9.24e-02	-9.26e-02	3.865	0.5823	5.54e-09
9	5.38e-09	5.98e-13	-9.24e-02	-9.25e-02	3.978	0.2157	8.79e-09
10	2.09e-09	2.32e-13	-9.24e-02	-9.25e-02	4.375	0.6120	1.54e-09
11	1.15e-09	1.28e-13	-9.24e-02	-9.25e-02	4.623	0.4502	1.28e-09
12	4.97e-10	5.52e-14	-9.25e-02	-9.25e-02	4.989	0.5673	4.36e-10
13	8.19e-11	9.11e-15	-9.25e-02	-9.25e-02	5.706	0.8351	2.60e-11
14	3.33e-12	3.75e-16	-9.25e-02	-9.25e-02	6.973	0.9594	2.96e-13
15	7.67e-14	7.10e-17	-9.25e-02	-9.25e-02	8.243	0.9769	5.14e-15
16	9.12e-16	6.60e-17	-9.25e-02	-9.25e-02	9.778	0.9924	3.02e-17
17	6.21e-16	8.14e-17	-9.25e-02	-9.25e-02	12.045	0.9950	3.77e-19

optimization converged

counter	pri_inf	dual_inf	pri_obj	dual_obj	sigfig	alpha	nu

0	9.51e+01	7.21e-03	-1.32e-01	-5.00e+02	0.000	0.0000	1.00e+02
1	4.95e-01	3.75e-05	-1.30e-01	-9.07e+02	0.000	0.9948	2.24e-05
2	2.48e-03	1.88e-07	-9.21e-02	-1.68e+01	0.000	0.9950	2.07e-06
3	5.39e-05	4.08e-09	-9.92e-02	-4.65e-01	0.477	0.9782	6.62e-07
4	3.69e-06	2.80e-10	-1.29e-01	-1.67e-01	1.466	0.9314	1.44e-07
5	5.95e-07	4.51e-11	-1.32e-01	-1.38e-01	2.227	0.8390	8.51e-08
6	4.61e-08	3.50e-12	-1.32e-01	-1.33e-01	3.153	0.9225	3.38e-09
7	1.11e-08	8.38e-13	-1.32e-01	-1.33e-01	3.728	0.7601	3.95e-09
8	1.61e-09	1.22e-13	-1.32e-01	-1.32e-01	4.423	0.8542	3.82e-10
9	6.25e-10	4.74e-14	-1.32e-01	-1.32e-01	4.840	0.6124	5.70e-10
10	5.61e-10	4.25e-14	-1.32e-01	-1.32e-01	4.886	0.1026	1.29e-09
11	1.83e-10	1.39e-14	-1.32e-01	-1.32e-01	5.356	0.6739	1.37e-10
12	7.98e-11	6.06e-15	-1.32e-01	-1.32e-01	5.724	0.5635	8.51e-11
13	6.64e-11	5.03e-15	-1.32e-01	-1.32e-01	5.767	0.1681	1.43e-10
14	1.14e-11	8.76e-16	-1.32e-01	-1.32e-01	6.516	0.8277	4.90e-12
15	8.77e-13	8.91e-17	-1.32e-01	-1.32e-01	7.613	0.9233	1.70e-13
16	1.09e-14	4.84e-17	-1.32e-01	-1.32e-01	9.418	0.9876	3.51e-16
17	9.15e-16	6.13e-17	-1.32e-01	-1.32e-01	11.306	0.9939	1.33e-18
18	6.75e-16	5.09e-17	-1.32e-01	-1.32e-01	13.603	0.9950	1.16e-20

optimization converged

counter	pri_inf	dual_inf	pri_obj	dual_obj	sigfig	alpha	nu

0	9.51e+01	7.56e-04	-2.61e+00	-5.03e+02	0.000	0.0000	1.00e+02
1	5.57e-01	4.43e-06	-2.59e+00	-9.09e+02	0.000	0.9941	2.83e-05

2	1.10e-02	8.77e-08	-2.14e+00	-3.32e+01	0.000	0.9802	3.30e-05
3	2.35e-03	1.87e-08	-2.29e+00	-8.96e+00	0.000	0.7872	1.24e-04
4	1.60e-04	1.27e-09	-2.53e+00	-3.09e+00	0.802	0.9319	2.60e-06
5	4.57e-05	3.63e-10	-2.57e+00	-2.74e+00	1.323	0.7146	3.96e-06
6	7.91e-06	6.29e-11	-2.60e+00	-2.64e+00	1.978	0.8268	4.35e-07
7	4.25e-07	3.38e-12	-2.61e+00	-2.61e+00	3.005	0.9463	9.11e-09
8	7.20e-08	5.72e-13	-2.61e+00	-2.61e+00	3.752	0.8305	8.73e-09
9	3.60e-10	2.87e-15	-2.61e+00	-2.61e+00	5.734	0.9950	1.32e-12
10	1.80e-12	1.92e-17	-2.61e+00	-2.61e+00	8.033	0.9950	1.38e-14
11	9.16e-15	1.12e-17	-2.61e+00	-2.61e+00	10.331	0.9950	6.92e-17
12	6.03e-16	1.64e-17	-2.61e+00	-2.61e+00	12.628	0.9950	3.49e-19

optimization converged							
counter	pri_inf	dual_inf	pri_obj	dual_obj	sigfig	alpha	nu

0	9.51e+01	1.75e-01	-3.56e-03	-5.00e+02	0.000	0.0000	1.00e+02
1	8.61e-01	1.58e-03	2.94e-04	-9.05e+02	0.000	0.9909	6.78e-05
2	4.30e-03	7.92e-06	2.33e-01	-2.28e+01	0.000	0.9950	2.21e-06
3	1.54e-04	2.84e-07	2.04e-01	-6.29e-01	0.160	0.9642	2.49e-06
4	1.85e-06	3.41e-09	3.16e-02	-5.18e-02	1.093	0.9880	9.95e-09
5	1.47e-07	2.71e-10	2.61e-03	-1.19e-02	1.841	0.9204	4.43e-08
6	2.11e-08	3.88e-11	-2.72e-03	-5.83e-03	2.509	0.8569	2.51e-08
7	1.06e-10	1.94e-13	-3.94e-03	-4.38e-03	3.363	0.9950	6.42e-12
8	1.17e-12	2.14e-15	-4.09e-03	-4.15e-03	4.227	0.9889	4.42e-12
9	6.14e-13	1.12e-15	-4.11e-03	-4.14e-03	4.526	0.4743	1.50e-09
10	2.87e-13	5.31e-16	-4.12e-03	-4.13e-03	4.875	0.5318	5.91e-10
11	1.50e-15	6.70e-17	-4.13e-03	-4.13e-03	6.173	0.9950	2.77e-14
12	7.94e-16	4.35e-17	-4.13e-03	-4.13e-03	8.075	0.9924	3.22e-15
13	8.60e-16	4.61e-17	-4.13e-03	-4.13e-03	10.298	0.9949	1.79e-17
14	7.99e-16	6.75e-17	-4.13e-03	-4.13e-03	12.596	0.9950	1.05e-19

optimization converged							
counter	pri_inf	dual_inf	pri_obj	dual_obj	sigfig	alpha	nu

0	9.51e+01	7.88e-04	-1.92e+00	-5.02e+02	0.000	0.0000	1.00e+02
1	4.96e-01	4.11e-06	-1.92e+00	-9.08e+02	0.000	0.9948	2.25e-05
2	7.07e-03	5.86e-08	-1.65e+00	-2.67e+01	0.000	0.9857	1.68e-05
3	1.28e-03	1.06e-08	-1.74e+00	-6.28e+00	0.000	0.8197	7.07e-05
4	1.47e-04	1.21e-09	-1.88e+00	-2.45e+00	0.708	0.8851	5.07e-06
5	2.14e-05	1.77e-10	-1.91e+00	-2.01e+00	1.482	0.8540	1.02e-06
6	9.88e-06	8.19e-11	-1.92e+00	-1.97e+00	1.813	0.5381	1.84e-06
7	4.39e-07	3.64e-12	-1.92e+00	-1.93e+00	2.931	0.9555	7.41e-09
8	3.82e-07	3.16e-12	-1.92e+00	-1.93e+00	2.992	0.1308	2.53e-07
9	1.12e-07	9.30e-13	-1.92e+00	-1.92e+00	3.493	0.7061	2.25e-08
10	3.24e-08	2.69e-13	-1.92e+00	-1.92e+00	4.003	0.7111	6.85e-09
11	1.46e-08	1.21e-13	-1.92e+00	-1.92e+00	4.350	0.5486	5.32e-09

12	5.44e-09	4.50e-14	-1.92e+00	-1.92e+00	4.790	0.6287	1.60e-09
13	1.37e-09	1.14e-14	-1.92e+00	-1.92e+00	5.376	0.7474	2.62e-10
14	2.90e-11	2.47e-16	-1.92e+00	-1.92e+00	6.800	0.9789	4.55e-13
15	4.73e-13	2.02e-17	-1.92e+00	-1.92e+00	8.042	0.9837	1.02e-14
16	4.59e-15	1.76e-17	-1.92e+00	-1.92e+00	9.185	0.9903	2.08e-16
17	6.66e-16	1.31e-17	-1.92e+00	-1.92e+00	11.094	0.9950	3.95e-18
18	6.44e-16	1.83e-17	-1.92e+00	-1.92e+00	13.399	0.9950	4.87e-20

optimization converged

counter	pri_inf	dual_inf	pri_obj	dual_obj	sigfig	alpha	nu

0	9.51e+01	1.17e-02	-8.88e-02	-5.00e+02	0.000	0.0000	1.00e+02
1	5.53e-01	6.80e-05	-8.49e-02	-9.06e+02	0.000	0.9942	2.80e-05
2	2.77e-03	3.40e-07	-2.34e-02	-1.78e+01	0.000	0.9950	2.09e-06
3	6.95e-05	8.54e-09	-3.43e-02	-4.84e-01	0.361	0.9749	9.41e-07
4	4.51e-06	5.54e-10	-8.30e-02	-1.35e-01	1.320	0.9351	1.58e-07
5	1.72e-06	2.11e-10	-8.71e-02	-1.06e-01	1.754	0.6193	6.66e-07
6	1.26e-06	1.55e-10	-8.87e-02	-1.03e-01	1.886	0.2671	9.77e-07
7	3.11e-07	3.82e-11	-8.99e-02	-9.34e-02	2.485	0.7531	7.46e-08
8	3.58e-08	4.40e-12	-9.06e-02	-9.13e-02	3.204	0.8847	4.01e-09
9	1.70e-08	2.09e-12	-9.07e-02	-9.10e-02	3.542	0.5262	1.38e-08
10	4.64e-09	5.70e-13	-9.08e-02	-9.09e-02	4.071	0.7266	2.04e-09
11	5.13e-10	6.30e-14	-9.09e-02	-9.09e-02	4.877	0.8895	9.54e-11
12	5.89e-11	7.24e-15	-9.09e-02	-9.09e-02	5.725	0.8852	1.61e-11
13	6.00e-13	1.05e-16	-9.09e-02	-9.09e-02	7.439	0.9898	1.77e-14
14	8.98e-14	6.80e-17	-9.09e-02	-9.09e-02	8.250	0.8504	7.54e-14
15	5.96e-15	1.08e-16	-9.09e-02	-9.09e-02	9.185	0.9354	2.14e-15
16	7.16e-16	1.16e-16	-9.09e-02	-9.09e-02	10.485	0.9897	6.28e-18
17	6.23e-16	8.42e-17	-9.09e-02	-9.09e-02	12.119	0.9945	9.10e-20

optimization converged

counter	pri_inf	dual_inf	pri_obj	dual_obj	sigfig	alpha	nu

0	9.51e+01	1.72e-03	-1.26e+00	-5.01e+02	0.000	0.0000	1.00e+02
1	5.83e-01	1.05e-05	-1.23e+00	-9.07e+02	0.000	0.9939	3.10e-05
2	9.03e-03	1.63e-07	-1.03e+00	-2.88e+01	0.000	0.9845	2.03e-05
3	7.50e-04	1.35e-08	-1.06e+00	-3.39e+00	0.000	0.9169	1.64e-05
4	3.65e-05	6.58e-10	-1.25e+00	-1.61e+00	0.797	0.9514	4.60e-07
5	9.56e-06	1.73e-10	-1.25e+00	-1.35e+00	1.379	0.7377	2.14e-06
6	2.18e-06	3.94e-11	-1.25e+00	-1.28e+00	2.022	0.7717	4.23e-07
7	1.60e-06	2.88e-11	-1.26e+00	-1.28e+00	2.110	0.2688	1.09e-06
8	1.17e-07	2.12e-12	-1.26e+00	-1.26e+00	3.107	0.9264	7.96e-09
9	3.39e-08	6.12e-13	-1.26e+00	-1.26e+00	3.616	0.7111	1.29e-08
10	7.42e-10	1.34e-14	-1.26e+00	-1.26e+00	5.083	0.9781	2.17e-11
11	3.73e-12	6.55e-17	-1.26e+00	-1.26e+00	7.371	0.9950	3.91e-14
12	1.88e-14	2.19e-17	-1.26e+00	-1.26e+00	9.668	0.9950	1.99e-16

13	6.66e-16	1.96e-17	-1.26e+00	-1.26e+00	11.966	0.9950	1.00e-18
14	4.99e-16	2.12e-17	-1.26e+00	-1.26e+00	14.318	0.9950	5.07e-21

optimization converged							
counter	pri_inf	dual_inf	pri_obj	dual_obj	sigfig	alpha	nu

0	9.51e+01	5.20e-02	-1.79e-02	-5.00e+02	0.000	0.0000	1.00e+02
1	7.54e-01	4.12e-04	-1.52e-02	-9.05e+02	0.000	0.9921	5.20e-05
2	3.77e-03	2.06e-06	1.40e-01	-2.11e+01	0.000	0.9950	2.17e-06
3	1.21e-04	6.58e-08	1.18e-01	-5.66e-01	0.213	0.9680	1.83e-06
4	1.22e-06	6.66e-10	-1.79e-03	-6.57e-02	1.195	0.9899	5.81e-09
5	3.86e-08	2.11e-11	-1.72e-02	-2.56e-02	2.085	0.9683	5.32e-09
6	1.89e-09	1.03e-12	-1.82e-02	-1.90e-02	3.135	0.9511	1.66e-09
7	4.19e-10	2.29e-13	-1.83e-02	-1.84e-02	3.769	0.7777	3.17e-09
8	6.57e-11	3.59e-14	-1.83e-02	-1.83e-02	4.524	0.8433	3.62e-10
9	2.23e-11	1.22e-14	-1.83e-02	-1.83e-02	4.931	0.6610	3.08e-10
10	8.30e-12	4.53e-15	-1.83e-02	-1.83e-02	5.344	0.6275	1.47e-10
11	2.93e-13	1.94e-16	-1.83e-02	-1.83e-02	6.455	0.9647	4.78e-13
12	4.81e-15	8.92e-17	-1.83e-02	-1.83e-02	8.015	0.9834	8.11e-15
13	6.88e-16	1.29e-16	-1.83e-02	-1.83e-02	9.818	0.9937	3.28e-17
14	9.20e-16	1.68e-16	-1.83e-02	-1.83e-02	12.057	0.9950	3.21e-19

optimization converged							
counter	pri_inf	dual_inf	pri_obj	dual_obj	sigfig	alpha	nu

0	9.51e+01	1.00e-03	-2.85e+00	-5.03e+02	0.000	0.0000	1.00e+02
1	8.53e-01	8.99e-06	-2.80e+00	-9.07e+02	0.000	0.9910	6.65e-05
2	3.16e-02	3.33e-07	-2.19e+00	-5.33e+01	0.000	0.9630	1.22e-04
3	6.94e-03	7.31e-08	-2.37e+00	-1.38e+01	0.000	0.7804	2.22e-04
4	3.55e-05	3.74e-10	-2.76e+00	-3.07e+00	1.079	0.9949	2.51e-08
5	1.09e-05	1.15e-10	-2.78e+00	-2.87e+00	1.596	0.6919	2.61e-06
6	7.22e-06	7.61e-11	-2.80e+00	-2.86e+00	1.778	0.3395	3.91e-06
7	2.27e-06	2.39e-11	-2.81e+00	-2.83e+00	2.260	0.6858	5.48e-07
8	9.35e-07	9.86e-12	-2.82e+00	-2.83e+00	2.648	0.5880	3.17e-07
9	9.16e-09	9.66e-14	-2.82e+00	-2.82e+00	4.409	0.9902	6.83e-11
10	4.60e-11	4.79e-16	-2.82e+00	-2.82e+00	6.704	0.9950	3.11e-13
11	2.30e-13	1.73e-17	-2.82e+00	-2.82e+00	9.002	0.9950	1.56e-15
12	1.29e-15	1.16e-17	-2.82e+00	-2.82e+00	11.299	0.9950	7.88e-18
13	5.13e-16	1.20e-17	-2.82e+00	-2.82e+00	13.577	0.9950	3.97e-20

optimization converged							
counter	pri_inf	dual_inf	pri_obj	dual_obj	sigfig	alpha	nu

0	9.51e+01	2.42e-03	-3.65e-01	-5.00e+02	0.000	0.0000	1.00e+02
1	4.76e-01	1.21e-05	-3.65e-01	-9.07e+02	0.000	0.9950	2.07e-05
2	2.38e-03	6.05e-08	-3.65e-01	-1.67e+01	0.000	0.9950	2.06e-06

3		1.49e-05		3.80e-10		-3.65e-01		-4.69e-01		1.117		0.9937		5.36e-08
4		7.23e-07		1.84e-11		-3.65e-01		-3.70e-01		2.424		0.9515		2.04e-08
5		6.35e-08		1.61e-12		-3.65e-01		-3.66e-01		3.392		0.9122		3.33e-09
6		1.15e-08		2.94e-13		-3.65e-01		-3.66e-01		4.081		0.8180		1.57e-09
7		4.11e-09		1.05e-13		-3.65e-01		-3.65e-01		4.534		0.6439		1.27e-09
8		4.63e-10		1.18e-14		-3.65e-01		-3.65e-01		5.440		0.8875		4.26e-11
9		1.64e-10		4.17e-15		-3.65e-01		-3.65e-01		5.877		0.6459		5.49e-11
10		7.48e-11		1.90e-15		-3.65e-01		-3.65e-01		6.216		0.5434		3.40e-11
11		3.12e-11		8.06e-16		-3.65e-01		-3.65e-01		6.594		0.5826		1.29e-11
12		1.07e-11		2.70e-16		-3.65e-01		-3.65e-01		7.045		0.6585		3.57e-12
13		2.97e-12		8.97e-17		-3.65e-01		-3.65e-01		7.595		0.7214		8.31e-13
14		7.94e-13		4.44e-17		-3.65e-01		-3.65e-01		8.156		0.7329		2.15e-13
15		1.05e-13		4.12e-17		-3.65e-01		-3.65e-01		9.018		0.8684		1.40e-14
16		1.80e-15		4.74e-17		-3.65e-01		-3.65e-01		10.750		0.9843		2.68e-17
17		7.49e-16		5.11e-17		-3.65e-01		-3.65e-01		13.039		0.9950		5.03e-20

optimization converged

counter		pri_inf		dual_inf		pri_obj		dual_obj		sigfig		alpha		nu

0		9.51e+01		1.83e-01		-3.11e-03		-5.00e+02		0.000		0.0000		1.00e+02
1		8.84e-01		1.71e-03		4.82e-04		-9.05e+02		0.000		0.9907		7.16e-05
2		4.42e-03		8.53e-06		2.41e-01		-2.32e+01		0.000		0.9950		2.22e-06
3		1.59e-04		3.06e-07		2.11e-01		-6.37e-01		0.155		0.9641		2.54e-06
4		1.94e-06		3.75e-09		3.40e-02		-5.11e-02		1.085		0.9878		1.05e-08
5		1.64e-07		3.15e-10		4.05e-03		-1.08e-02		1.829		0.9158		5.06e-08
6		1.22e-08		2.35e-11		-2.02e-03		-4.67e-03		2.578		0.9254		6.94e-09
7		2.64e-09		5.08e-12		-3.02e-03		-3.70e-03		3.167		0.7840		1.06e-08
8		3.11e-10		6.01e-13		-3.33e-03		-3.46e-03		3.870		0.8819		8.05e-10
9		4.06e-12		7.85e-15		-3.40e-03		-3.42e-03		4.769		0.9870		1.91e-12
10		3.79e-14		8.35e-17		-3.42e-03		-3.42e-03		5.931		0.9907		1.23e-13
11		7.36e-16		7.84e-17		-3.42e-03		-3.42e-03		8.014		0.9950		2.43e-15
12		7.34e-16		1.38e-16		-3.42e-03		-3.42e-03		10.313		0.9950		2.01e-17
13		6.14e-16		7.23e-17		-3.42e-03		-3.42e-03		12.611		0.9950		1.01e-19

optimization converged

counter		pri_inf		dual_inf		pri_obj		dual_obj		sigfig		alpha		nu

0		9.51e+01		2.95e-02		-3.47e-02		-5.00e+02		0.000		0.0000		1.00e+02
1		6.62e-01		2.06e-04		-3.20e-02		-9.06e+02		0.000		0.9930		4.01e-05
2		3.31e-03		1.03e-06		9.00e-02		-1.96e+01		0.000		0.9950		2.13e-06
3		9.95e-05		3.09e-08		7.18e-02		-5.25e-01		0.255		0.9699		1.49e-06
4		1.83e-06		5.70e-10		-2.24e-02		-7.89e-02		1.258		0.9816		1.68e-08
5		5.79e-07		1.80e-10		-3.15e-02		-4.93e-02		1.763		0.6840		4.94e-07
6		1.12e-07		3.48e-11		-3.40e-02		-3.77e-02		2.449		0.8066		5.70e-08
7		2.20e-09		6.82e-13		-3.49e-02		-3.55e-02		3.258		0.9804		1.17e-10
8		4.54e-10		1.41e-13		-3.50e-02		-3.51e-02		3.908		0.7932		2.10e-09

9	1.45e-10	4.50e-14	-3.50e-02	-3.50e-02	4.410	0.6805	1.15e-09
10	7.35e-11	2.28e-14	-3.50e-02	-3.50e-02	4.714	0.4935	9.37e-10
11	3.73e-11	1.16e-14	-3.50e-02	-3.50e-02	5.005	0.4920	4.69e-10
12	1.43e-12	4.57e-16	-3.50e-02	-3.50e-02	5.817	0.9616	1.25e-12
13	4.02e-13	1.72e-16	-3.50e-02	-3.50e-02	6.369	0.7198	1.08e-11
14	5.38e-14	9.91e-17	-3.50e-02	-3.50e-02	7.233	0.8663	6.71e-13
15	1.03e-15	1.17e-16	-3.50e-02	-3.50e-02	9.137	0.9915	3.61e-16
16	9.02e-16	8.01e-17	-3.50e-02	-3.50e-02	11.385	0.9949	1.61e-18
17	6.53e-16	1.23e-16	-3.50e-02	-3.50e-02	13.684	0.9950	8.81e-21

optimization converged							
counter	pri_inf	dual_inf	pri_obj	dual_obj	sigfig	alpha	nu

0	9.51e+01	7.52e-03	-1.46e-01	-5.00e+02	0.000	0.0000	1.00e+02
1	5.57e-01	4.41e-05	-1.42e-01	-9.06e+02	0.000	0.9941	2.84e-05
2	2.79e-03	2.20e-07	-9.48e-02	-1.80e+01	0.000	0.9950	2.09e-06
3	5.92e-05	4.68e-09	-1.02e-01	-4.85e-01	0.460	0.9787	6.76e-07
4	5.87e-06	4.64e-10	-1.36e-01	-1.86e-01	1.358	0.9009	3.16e-07
5	3.07e-06	2.42e-10	-1.45e-01	-1.74e-01	1.597	0.4774	1.24e-06
6	2.22e-07	1.75e-11	-1.47e-01	-1.50e-01	2.565	0.9276	1.27e-08
7	1.38e-07	1.09e-11	-1.48e-01	-1.49e-01	2.784	0.3800	1.11e-07
8	3.35e-08	2.65e-12	-1.48e-01	-1.48e-01	3.330	0.7566	9.66e-09
9	1.26e-08	9.92e-13	-1.48e-01	-1.48e-01	3.761	0.6252	6.69e-09
10	7.69e-09	6.08e-13	-1.48e-01	-1.48e-01	3.988	0.3873	6.93e-09
11	4.37e-09	3.45e-13	-1.48e-01	-1.48e-01	4.226	0.4318	3.50e-09
12	1.68e-09	1.33e-13	-1.48e-01	-1.48e-01	4.634	0.6149	8.98e-10
13	1.51e-09	1.20e-13	-1.48e-01	-1.48e-01	4.675	0.1015	2.11e-09
14	1.78e-10	1.41e-14	-1.48e-01	-1.48e-01	5.463	0.8821	2.85e-11
15	2.60e-11	2.07e-15	-1.48e-01	-1.48e-01	6.273	0.8542	7.13e-12
16	2.96e-13	6.69e-17	-1.48e-01	-1.48e-01	8.086	0.9886	6.56e-15
17	5.81e-15	7.41e-17	-1.48e-01	-1.48e-01	9.675	0.9803	3.04e-16
18	7.77e-16	6.43e-17	-1.48e-01	-1.48e-01	11.429	0.9933	9.06e-19
19	6.39e-16	9.70e-17	-1.48e-01	-1.48e-01	13.716	0.9950	8.85e-21

optimization converged							
counter	pri_inf	dual_inf	pri_obj	dual_obj	sigfig	alpha	nu

0	9.51e+01	1.64e-01	-3.93e-03	-5.00e+02	0.000	0.0000	1.00e+02
1	8.55e-01	1.48e-03	-2.54e-04	-9.05e+02	0.000	0.9910	6.68e-05
2	4.27e-03	7.39e-06	2.27e-01	-2.27e+01	0.000	0.9950	2.21e-06
3	1.52e-04	2.63e-07	1.98e-01	-6.24e-01	0.163	0.9644	2.44e-06
4	1.81e-06	3.13e-09	2.94e-02	-5.29e-02	1.097	0.9881	9.64e-09
5	1.41e-07	2.43e-10	1.62e-03	-1.25e-02	1.851	0.9222	4.17e-08
6	5.72e-08	9.88e-11	-2.25e-03	-7.89e-03	2.250	0.5935	2.08e-07
7	7.11e-10	1.23e-12	-4.23e-03	-5.02e-03	3.100	0.9876	7.22e-11
8	1.13e-11	1.96e-14	-4.44e-03	-4.54e-03	4.005	0.9841	1.68e-11

9		3.40e-12		5.86e-15		-4.46e-03		-4.49e-03		4.538		0.6998		7.81e-10
10		2.56e-12		4.41e-15		-4.46e-03		-4.49e-03		4.668		0.2474		1.57e-09
11		4.83e-13		8.47e-16		-4.47e-03		-4.47e-03		5.297		0.8113		6.57e-11
12		1.86e-13		3.23e-16		-4.47e-03		-4.47e-03		5.715		0.6148		6.68e-11
13		6.03e-14		1.34e-16		-4.47e-03		-4.47e-03		6.208		0.6756		1.79e-11
14		3.05e-15		7.68e-17		-4.47e-03		-4.47e-03		7.484		0.9513		1.23e-13
15		7.58e-16		9.61e-17		-4.47e-03		-4.47e-03		9.741		0.9948		7.28e-17
16		8.98e-16		8.09e-17		-4.47e-03		-4.47e-03		12.039		0.9950		3.77e-19

optimization converged

counter		pri_inf		dual_inf		pri_obj		dual_obj		sigfig		alpha		nu

0		9.51e+01		1.45e-03		-9.01e-01		-5.01e+02		0.000		0.0000		1.00e+02
1		4.76e-01		7.27e-06		-8.92e-01		-9.07e+02		0.000		0.9950		2.07e-05
2		3.72e-03		5.69e-08		-8.30e-01		-1.97e+01		0.000		0.9922		5.05e-06
3		2.41e-04		3.68e-09		-8.42e-01		-2.07e+00		0.176		0.9352		6.70e-06
4		3.61e-05		5.52e-10		-8.95e-01		-1.11e+00		0.954		0.8502		2.34e-06
5		2.49e-05		3.81e-10		-9.00e-01		-1.04e+00		1.120		0.3090		9.46e-06
6		2.85e-06		4.36e-11		-9.05e-01		-9.22e-01		2.033		0.8856		1.59e-07
7		3.22e-07		4.93e-12		-9.06e-01		-9.08e-01		2.916		0.8870		1.90e-08
8		1.73e-07		2.65e-12		-9.06e-01		-9.07e-01		3.188		0.4631		6.09e-08
9		2.98e-09		4.56e-14		-9.06e-01		-9.06e-01		4.202		0.9828		3.05e-11
10		3.26e-11		4.89e-16		-9.06e-01		-9.06e-01		5.401		0.9891		1.18e-12
11		1.90e-13		4.20e-17		-9.06e-01		-9.06e-01		7.469		0.9942		2.13e-14
12		1.09e-15		3.31e-17		-9.06e-01		-9.06e-01		9.767		0.9950		1.34e-16
13		7.91e-16		2.48e-17		-9.06e-01		-9.06e-01		12.065		0.9950		6.74e-19

optimization converged

counter		pri_inf		dual_inf		pri_obj		dual_obj		sigfig		alpha		nu

0		9.51e+01		3.42e-02		-3.28e-02		-5.00e+02		0.000		0.0000		1.00e+02
1		6.36e-01		2.29e-04		-2.94e-02		-9.06e+02		0.000		0.9933		3.70e-05
2		3.18e-03		1.14e-06		9.96e-02		-1.92e+01		0.000		0.9950		2.12e-06
3		9.81e-05		3.52e-08		8.01e-02		-5.17e-01		0.257		0.9692		1.53e-06
4		1.53e-06		5.51e-10		-1.86e-02		-7.48e-02		1.258		0.9844		1.21e-08
5		1.34e-07		4.81e-11		-3.02e-02		-3.84e-02		2.099		0.9126		3.61e-08
6		6.85e-08		2.46e-11		-3.23e-02		-3.71e-02		2.331		0.4888		1.95e-07
7		1.56e-08		5.60e-12		-3.30e-02		-3.43e-02		2.931		0.7724		2.15e-08
8		1.33e-08		4.78e-12		-3.32e-02		-3.42e-02		2.986		0.1463		8.58e-08
9		1.56e-09		5.60e-13		-3.34e-02		-3.36e-02		3.747		0.8827		1.24e-09
10		5.68e-10		2.04e-13		-3.34e-02		-3.35e-02		4.196		0.6357		2.17e-09
11		3.11e-10		1.12e-13		-3.34e-02		-3.35e-02		4.444		0.4528		1.80e-09
12		2.65e-10		9.53e-14		-3.34e-02		-3.35e-02		4.520		0.1469		2.63e-09
13		1.12e-10		4.02e-14		-3.34e-02		-3.35e-02		4.793		0.5778		4.97e-10
14		5.87e-12		2.12e-15		-3.34e-02		-3.34e-02		5.984		0.9476		3.81e-12
15		3.20e-14		8.86e-17		-3.34e-02		-3.34e-02		8.153		0.9946		2.62e-15

16		7.19e-16		9.94e-17		-3.34e-02		-3.34e-02		10.451		0.9950		1.50e-17
17		7.52e-16		1.36e-16		-3.34e-02		-3.34e-02		12.749		0.9950		7.56e-20

optimization converged														
counter		pri_inf		dual_inf		pri_obj		dual_obj		sigfig		alpha		nu

0		9.51e+01		1.59e-03		-2.32e+00		-5.02e+02		0.000		0.0000		1.00e+02
1		5.81e-01		9.69e-06		-2.24e+00		-9.08e+02		0.000		0.9939		3.08e-05
2		1.60e-02		2.66e-07		-1.70e+00		-4.03e+01		0.000		0.9725		6.38e-05
3		1.92e-03		3.21e-08		-1.76e+00		-6.47e+00		0.000		0.8796		4.85e-05
4		5.04e-04		8.41e-09		-2.22e+00		-3.66e+00		0.350		0.7376		2.82e-05
5		3.26e-05		5.43e-10		-2.32e+00		-2.45e+00		1.412		0.9354		5.03e-07
6		5.97e-07		9.97e-12		-2.34e+00		-2.35e+00		2.520		0.9817		3.58e-09
7		1.14e-07		1.90e-12		-2.34e+00		-2.34e+00		3.242		0.8097		3.13e-08
8		1.02e-07		1.70e-12		-2.34e+00		-2.34e+00		3.290		0.1030		1.51e-07
9		1.00e-08		1.68e-13		-2.34e+00		-2.34e+00		4.194		0.9015		1.40e-09
10		9.23e-11		1.54e-15		-2.34e+00		-2.34e+00		6.201		0.9908		1.49e-12
11		4.62e-13		2.48e-17		-2.34e+00		-2.34e+00		8.496		0.9950		4.35e-15
12		2.55e-15		2.50e-17		-2.34e+00		-2.34e+00		10.793		0.9950		2.21e-17
13		6.90e-16		3.01e-17		-2.34e+00		-2.34e+00		13.084		0.9950		1.11e-19

optimization converged														
counter		pri_inf		dual_inf		pri_obj		dual_obj		sigfig		alpha		nu

0		9.51e+01		9.89e-04		-1.37e+00		-5.01e+02		0.000		0.0000		1.00e+02
1		4.76e-01		4.95e-06		-1.36e+00		-9.08e+02		0.000		0.9950		2.07e-05
2		4.91e-03		5.11e-08		-1.22e+00		-2.24e+01		0.000		0.9897		8.80e-06
3		6.93e-04		7.21e-09		-1.27e+00		-4.26e+00		0.000		0.8588		3.62e-05
4		5.89e-05		6.13e-10		-1.35e+00		-1.63e+00		0.918		0.9150		1.82e-06
5		2.04e-05		2.12e-10		-1.36e+00		-1.46e+00		1.373		0.6542		2.98e-06
6		1.01e-06		1.05e-11		-1.38e+00		-1.38e+00		2.466		0.9503		2.07e-08
7		5.32e-07		5.54e-12		-1.38e+00		-1.38e+00		2.733		0.4741		2.05e-07
8		4.23e-08		4.40e-13		-1.38e+00		-1.38e+00		3.590		0.9206		2.33e-09
9		8.00e-09		8.33e-14		-1.38e+00		-1.38e+00		4.302		0.8108		1.87e-09
10		6.69e-09		6.96e-14		-1.38e+00		-1.38e+00		4.377		0.1643		8.01e-09
11		3.30e-10		3.43e-15		-1.38e+00		-1.38e+00		5.507		0.9507		2.02e-11
12		1.87e-12		3.17e-17		-1.38e+00		-1.38e+00		7.687		0.9943		1.96e-14
13		9.26e-15		2.40e-17		-1.38e+00		-1.38e+00		9.985		0.9950		1.01e-16
14		7.14e-16		2.28e-17		-1.38e+00		-1.38e+00		12.282		0.9950		5.09e-19

optimization converged														
counter		pri_inf		dual_inf		pri_obj		dual_obj		sigfig		alpha		nu

0		9.51e+01		1.73e-01		-3.10e-03		-5.00e+02		0.000		0.0000		1.00e+02
1		8.81e-01		1.61e-03		4.40e-04		-9.05e+02		0.000		0.9907		7.11e-05
2		4.41e-03		8.03e-06		2.34e-01		-2.31e+01		0.000		0.9950		2.22e-06

3	1.58e-04	2.87e-07	2.05e-01	-6.38e-01	0.155	0.9642	2.52e-06
4	1.90e-06	3.47e-09	3.25e-02	-5.16e-02	1.089	0.9879	1.02e-08
5	1.76e-07	3.20e-10	3.70e-03	-1.15e-02	1.819	0.9078	6.01e-08
6	3.11e-08	5.68e-11	-1.63e-03	-5.20e-03	2.448	0.8228	4.08e-08
7	3.87e-09	7.05e-12	-3.04e-03	-3.78e-03	3.129	0.8759	4.65e-09
8	9.72e-11	1.77e-13	-3.33e-03	-3.44e-03	3.966	0.9748	3.92e-11
9	1.66e-11	3.03e-14	-3.37e-03	-3.39e-03	4.618	0.8290	2.70e-10
10	2.86e-12	5.22e-15	-3.38e-03	-3.39e-03	5.018	0.8279	6.11e-11
11	2.72e-14	7.34e-17	-3.38e-03	-3.38e-03	6.337	0.9905	7.20e-14
12	1.01e-15	6.33e-17	-3.38e-03	-3.38e-03	7.887	0.9906	3.36e-15
13	7.83e-16	5.26e-17	-3.38e-03	-3.38e-03	10.001	0.9948	2.97e-17
14	8.78e-16	7.66e-17	-3.38e-03	-3.38e-03	12.299	0.9950	2.07e-19

optimization converged

counter	pri_inf	dual_inf	pri_obj	dual_obj	sigfig	alpha	nu

0	9.51e+01	2.31e-03	-8.84e-01	-5.01e+02	0.000	0.0000	1.00e+02
1	7.45e-01	1.81e-05	-8.64e-01	-9.06e+02	0.000	0.9922	5.08e-05
2	1.09e-02	2.64e-07	-7.21e-01	-3.04e+01	0.000	0.9854	1.86e-05
3	7.47e-04	1.81e-08	-7.32e-01	-2.80e+00	0.000	0.9314	1.20e-05
4	3.28e-05	7.97e-10	-8.77e-01	-1.22e+00	0.744	0.9560	3.34e-07
5	2.13e-06	5.16e-11	-8.79e-01	-9.01e-01	1.925	0.9352	1.19e-07
6	1.40e-06	3.39e-11	-8.83e-01	-8.99e-01	2.073	0.3439	8.99e-07
7	3.60e-07	8.73e-12	-8.85e-01	-8.90e-01	2.618	0.7421	9.17e-08
8	8.37e-08	2.03e-12	-8.86e-01	-8.87e-01	3.214	0.7674	2.12e-08
9	1.17e-08	2.85e-13	-8.87e-01	-8.87e-01	3.979	0.8597	1.92e-09
10	6.39e-11	1.54e-15	-8.87e-01	-8.87e-01	6.145	0.9946	4.85e-13
11	3.20e-13	2.82e-17	-8.87e-01	-8.87e-01	8.443	0.9950	2.79e-15
12	1.84e-15	3.44e-17	-8.87e-01	-8.87e-01	10.741	0.9950	1.41e-17
13	6.57e-16	3.93e-17	-8.87e-01	-8.87e-01	13.039	0.9950	7.09e-20

optimization converged

counter	pri_inf	dual_inf	pri_obj	dual_obj	sigfig	alpha	nu

0	9.51e+01	4.67e-03	-1.94e-01	-5.00e+02	0.000	0.0000	1.00e+02
1	4.76e-01	2.34e-05	-1.93e-01	-9.07e+02	0.000	0.9950	2.07e-05
2	2.38e-03	1.17e-07	-1.75e-01	-1.65e+01	0.000	0.9950	2.06e-06
3	4.11e-05	2.02e-09	-1.78e-01	-4.63e-01	0.617	0.9827	4.08e-07
4	2.35e-06	1.16e-10	-1.93e-01	-2.15e-01	1.717	0.9427	7.79e-08
5	2.82e-07	1.39e-11	-1.94e-01	-1.96e-01	2.606	0.8799	2.78e-08
6	5.10e-08	2.51e-12	-1.94e-01	-1.94e-01	3.328	0.8193	8.26e-09
7	3.05e-08	1.50e-12	-1.94e-01	-1.94e-01	3.552	0.4028	1.85e-08
8	6.62e-09	3.25e-13	-1.94e-01	-1.94e-01	4.166	0.7827	1.36e-09
9	1.87e-09	9.17e-14	-1.94e-01	-1.94e-01	4.706	0.7180	5.64e-10
10	5.20e-10	2.56e-14	-1.94e-01	-1.94e-01	5.217	0.7214	1.59e-10
11	1.15e-10	5.64e-15	-1.94e-01	-1.94e-01	5.831	0.7790	3.04e-11

12		2.04e-12		1.14e-16		-1.94e-01		-1.94e-01		7.037		0.9822		4.62e-14
13		2.81e-13		6.86e-17		-1.94e-01		-1.94e-01		7.873		0.8623		1.76e-13
14		1.72e-15		5.99e-17		-1.94e-01		-1.94e-01		9.577		0.9950		3.31e-17
15		7.37e-16		6.26e-17		-1.94e-01		-1.94e-01		11.810		0.9949		6.90e-19
16		5.58e-16		6.20e-17		-1.94e-01		-1.94e-01		14.124		0.9950		3.83e-21

optimization converged														
counter		pri_inf		dual_inf		pri_obj		dual_obj		sigfig		alpha		nu

0		9.51e+01		6.98e-02		-1.49e-02		-5.00e+02		0.000		0.0000		1.00e+02
1		7.54e-01		5.53e-04		-1.04e-02		-9.05e+02		0.000		0.9921		5.20e-05
2		3.77e-03		2.77e-06		1.59e-01		-2.11e+01		0.000		0.9950		2.17e-06
3		1.24e-04		9.11e-08		1.35e-01		-5.70e-01		0.207		0.9671		1.94e-06
4		1.33e-06		9.72e-10		5.26e-03		-6.24e-02		1.172		0.9893		6.66e-09
5		1.65e-07		1.21e-10		-1.22e-02		-2.53e-02		1.888		0.8757		8.84e-08
6		1.33e-07		9.78e-11		-1.31e-02		-2.33e-02		1.995		0.1908		8.25e-07
7		3.06e-08		2.25e-11		-1.49e-02		-1.75e-02		2.601		0.7701		4.67e-08
8		1.10e-08		8.05e-12		-1.55e-02		-1.65e-02		2.994		0.6418		2.88e-08
9		5.93e-09		4.35e-12		-1.57e-02		-1.62e-02		3.261		0.4599		2.75e-08
10		3.16e-09		2.32e-12		-1.58e-02		-1.61e-02		3.522		0.4673		1.44e-08
11		3.47e-10		2.55e-13		-1.60e-02		-1.60e-02		4.370		0.8900		3.12e-10
12		5.76e-12		4.24e-15		-1.60e-02		-1.60e-02		5.800		0.9834		9.86e-13
13		3.51e-14		2.09e-16		-1.60e-02		-1.60e-02		7.939		0.9939		4.98e-15
14		7.18e-16		3.05e-16		-1.60e-02		-1.60e-02		10.237		0.9950		2.42e-17
15		5.80e-16		2.87e-16		-1.60e-02		-1.60e-02		12.534		0.9950		1.22e-19

optimization converged														
counter		pri_inf		dual_inf		pri_obj		dual_obj		sigfig		alpha		nu

0		9.51e+01		2.61e-03		-5.27e-01		-5.00e+02		0.000		0.0000		1.00e+02
1		6.42e-01		1.76e-05		-5.21e-01		-9.06e+02		0.000		0.9933		3.76e-05
2		5.50e-03		1.51e-07		-4.73e-01		-2.30e+01		0.000		0.9914		6.25e-06
3		2.44e-04		6.69e-09		-4.76e-01		-1.48e+00		0.166		0.9557		3.74e-06
4		5.47e-06		1.50e-10		-5.25e-01		-6.24e-01		1.189		0.9776		4.21e-08
5		3.54e-07		9.73e-12		-5.27e-01		-5.34e-01		2.346		0.9351		3.47e-08
6		2.24e-08		6.14e-13		-5.28e-01		-5.29e-01		3.158		0.9369		2.29e-09
7		3.32e-09		9.11e-14		-5.28e-01		-5.28e-01		3.962		0.8516		1.99e-09
8		2.64e-09		7.24e-14		-5.28e-01		-5.28e-01		4.058		0.2057		1.01e-08
9		5.72e-10		1.57e-14		-5.28e-01		-5.28e-01		4.655		0.7831		5.41e-10
10		2.23e-10		6.12e-15		-5.28e-01		-5.28e-01		5.063		0.6099		4.57e-10
11		7.60e-11		2.08e-15		-5.28e-01		-5.28e-01		5.526		0.6594		1.35e-10
12		5.10e-12		1.47e-16		-5.28e-01		-5.28e-01		6.640		0.9329		1.71e-12
13		2.83e-14		2.47e-17		-5.28e-01		-5.28e-01		8.843		0.9945		8.90e-16
14		1.06e-15		3.41e-17		-5.28e-01		-5.28e-01		11.141		0.9950		4.54e-18
15		7.03e-16		3.22e-17		-5.28e-01		-5.28e-01		13.436		0.9950		2.29e-20

optimization converged

counter	pri_inf	dual_inf	pri_obj	dual_obj	sigfig	alpha	nu
0	9.51e+01	8.44e-04	-2.22e+00	-5.02e+02	0.000	0.0000	1.00e+02
1	5.08e-01	4.51e-06	-2.20e+00	-9.09e+02	0.000	0.9947	2.36e-05
2	9.42e-03	8.35e-08	-1.87e+00	-3.09e+01	0.000	0.9815	2.86e-05
3	1.89e-03	1.68e-08	-1.99e+00	-7.87e+00	0.000	0.7994	1.02e-04
4	3.97e-05	3.52e-10	-2.18e+00	-2.40e+00	1.157	0.9790	2.16e-07
5	1.83e-05	1.63e-10	-2.22e+00	-2.33e+00	1.444	0.5376	4.27e-06
6	3.51e-06	3.11e-11	-2.23e+00	-2.25e+00	2.107	0.8088	3.62e-07
7	8.64e-08	7.66e-13	-2.23e+00	-2.23e+00	3.192	0.9754	1.27e-09
8	3.02e-08	2.68e-13	-2.23e+00	-2.23e+00	3.662	0.6502	2.24e-08
9	2.32e-08	2.05e-13	-2.23e+00	-2.23e+00	3.752	0.2332	3.96e-08
10	1.10e-08	9.74e-14	-2.23e+00	-2.23e+00	4.080	0.5260	1.16e-08
11	1.07e-08	9.46e-14	-2.23e+00	-2.23e+00	4.086	0.0286	2.52e-08
12	3.73e-09	3.31e-14	-2.23e+00	-2.23e+00	4.486	0.6503	2.86e-09
13	6.10e-10	5.41e-15	-2.23e+00	-2.23e+00	5.238	0.8365	2.40e-10
14	4.61e-10	4.09e-15	-2.23e+00	-2.23e+00	5.342	0.2434	1.02e-09
15	2.75e-11	2.47e-16	-2.23e+00	-2.23e+00	6.507	0.9404	4.37e-12
16	1.84e-13	1.66e-17	-2.23e+00	-2.23e+00	8.569	0.9933	3.71e-15
17	1.19e-15	1.64e-17	-2.23e+00	-2.23e+00	10.857	0.9950	1.81e-17
18	6.18e-16	2.39e-17	-2.23e+00	-2.23e+00	13.169	0.9950	9.29e-20

optimization converged

counter	pri_inf	dual_inf	pri_obj	dual_obj	sigfig	alpha	nu
0	9.51e+01	1.75e-01	-3.06e-03	-5.00e+02	0.000	0.0000	1.00e+02
1	8.83e-01	1.63e-03	4.40e-04	-9.05e+02	0.000	0.9907	7.13e-05
2	4.41e-03	8.14e-06	2.36e-01	-2.31e+01	0.000	0.9950	2.22e-06
3	1.58e-04	2.91e-07	2.06e-01	-6.39e-01	0.155	0.9642	2.53e-06
4	1.91e-06	3.52e-09	3.28e-02	-5.15e-02	1.088	0.9879	1.02e-08
5	2.01e-07	3.70e-10	3.97e-03	-1.20e-02	1.799	0.8951	7.82e-08
6	7.06e-09	1.30e-11	-2.21e-03	-4.69e-03	2.606	0.9648	1.64e-09
7	2.08e-09	3.83e-12	-3.03e-03	-3.83e-03	3.099	0.7060	1.87e-08
8	1.31e-10	2.42e-13	-3.32e-03	-3.46e-03	3.878	0.9369	2.66e-10
9	2.28e-12	4.19e-15	-3.37e-03	-3.39e-03	4.718	0.9826	3.33e-12
10	2.22e-14	7.50e-17	-3.38e-03	-3.38e-03	5.598	0.9903	1.49e-13
11	7.07e-15	5.87e-17	-3.38e-03	-3.38e-03	6.098	0.6800	2.27e-11
12	4.66e-15	5.75e-17	-3.38e-03	-3.38e-03	6.954	0.9950	1.66e-15
13	2.52e-15	7.81e-17	-3.38e-03	-3.38e-03	7.257	0.4762	2.79e-12
14	6.54e-16	8.21e-17	-3.38e-03	-3.38e-03	8.179	0.9631	6.28e-15
15	6.42e-16	4.06e-17	-3.38e-03	-3.38e-03	8.668	0.6778	6.04e-14
16	1.02e-15	7.95e-17	-3.38e-03	-3.38e-03	10.354	0.9813	6.25e-17
17	6.73e-16	5.34e-17	-3.38e-03	-3.38e-03	12.643	0.9950	9.18e-20

optimization converged

counter	pri_inf	dual_inf	pri_obj	dual_obj	sigfig	alpha	nu
0	9.51e+01	1.79e-03	-6.18e-01	-5.01e+02	0.000	0.0000	1.00e+02
1	5.11e-01	9.61e-06	-6.13e-01	-9.07e+02	0.000	0.9946	2.38e-05
2	3.77e-03	7.10e-08	-5.85e-01	-1.97e+01	0.000	0.9926	4.54e-06
3	2.53e-04	4.76e-09	-5.90e-01	-1.88e+00	0.092	0.9330	7.27e-06
4	2.67e-05	5.03e-10	-6.09e-01	-7.51e-01	1.054	0.8943	1.21e-06
5	7.30e-06	1.38e-10	-6.19e-01	-6.65e-01	1.547	0.7267	9.23e-07
6	3.80e-07	7.14e-12	-6.20e-01	-6.23e-01	2.774	0.9480	1.04e-08
7	2.03e-07	3.83e-12	-6.20e-01	-6.22e-01	3.045	0.4642	7.15e-08
8	8.60e-08	1.62e-12	-6.20e-01	-6.21e-01	3.410	0.5769	2.34e-08
9	1.75e-08	3.30e-13	-6.21e-01	-6.21e-01	4.053	0.7963	2.24e-09
10	7.56e-09	1.42e-13	-6.21e-01	-6.21e-01	4.416	0.5687	2.39e-09
11	1.58e-09	2.98e-14	-6.21e-01	-6.21e-01	5.093	0.7906	2.34e-10
12	3.10e-10	5.84e-15	-6.21e-01	-6.21e-01	5.729	0.8044	4.28e-11
13	2.22e-10	4.17e-15	-6.21e-01	-6.21e-01	5.875	0.2834	1.47e-10
14	5.35e-12	1.10e-16	-6.21e-01	-6.21e-01	7.348	0.9759	1.04e-13
15	2.50e-13	3.83e-17	-6.21e-01	-6.21e-01	8.563	0.9532	1.33e-14
16	4.57e-15	4.24e-17	-6.21e-01	-6.21e-01	10.067	0.9822	1.17e-16
17	7.41e-16	3.34e-17	-6.21e-01	-6.21e-01	11.929	0.9936	4.73e-19
18	6.32e-16	5.49e-17	-6.21e-01	-6.21e-01	14.307	0.9950	3.94e-21

optimization converged

counter	pri_inf	dual_inf	pri_obj	dual_obj	sigfig	alpha	nu
0	9.51e+01	1.57e-03	-1.82e+00	-5.02e+02	0.000	0.0000	1.00e+02
1	5.46e-01	8.99e-06	-1.77e+00	-9.08e+02	0.000	0.9943	2.73e-05
2	1.14e-02	1.88e-07	-1.41e+00	-3.35e+01	0.000	0.9791	3.67e-05
3	1.21e-03	1.99e-08	-1.45e+00	-4.88e+00	0.000	0.8943	3.09e-05
4	2.39e-04	3.94e-09	-1.77e+00	-2.65e+00	0.497	0.8022	1.15e-05
5	1.14e-05	1.87e-10	-1.81e+00	-1.87e+00	1.673	0.9525	1.67e-07
6	4.95e-06	8.16e-11	-1.81e+00	-1.84e+00	2.042	0.5645	1.01e-06
7	1.77e-06	2.91e-11	-1.82e+00	-1.83e+00	2.490	0.6428	2.88e-07
8	7.32e-07	1.20e-11	-1.82e+00	-1.82e+00	2.865	0.5864	1.39e-07
9	3.15e-07	5.19e-12	-1.82e+00	-1.82e+00	3.238	0.5696	6.39e-08
10	9.21e-09	1.52e-13	-1.82e+00	-1.82e+00	4.622	0.9708	1.16e-10
11	4.60e-11	7.66e-16	-1.82e+00	-1.82e+00	6.907	0.9950	1.39e-13
12	2.30e-13	3.06e-17	-1.82e+00	-1.82e+00	9.204	0.9950	7.23e-16
13	1.34e-15	2.13e-17	-1.82e+00	-1.82e+00	11.502	0.9950	3.64e-18
14	7.07e-16	3.45e-17	-1.82e+00	-1.82e+00	13.803	0.9950	1.84e-20

optimization converged

counter	pri_inf	dual_inf	pri_obj	dual_obj	sigfig	alpha	nu
0	9.51e+01	1.49e-01	-5.09e-03	-5.00e+02	0.000	0.0000	1.00e+02
1	8.38e-01	1.31e-03	-1.24e-03	-9.05e+02	0.000	0.9912	6.43e-05

2	4.19e-03	6.55e-06	2.17e-01	-2.24e+01	0.000	0.9950	2.20e-06
3	1.48e-04	2.31e-07	1.89e-01	-6.16e-01	0.170	0.9648	2.36e-06
4	1.74e-06	2.71e-09	2.58e-02	-5.46e-02	1.106	0.9882	9.24e-09
5	1.34e-07	2.09e-10	-3.83e-04	-1.41e-02	1.863	0.9229	4.00e-08
6	4.94e-08	7.72e-11	-4.00e-03	-9.06e-03	2.298	0.6310	1.65e-07
7	8.85e-09	1.38e-11	-5.39e-03	-6.55e-03	2.936	0.8208	1.39e-08
8	2.21e-09	3.45e-12	-5.67e-03	-6.01e-03	3.468	0.7502	6.28e-09
9	9.39e-10	1.47e-12	-5.71e-03	-5.85e-03	3.859	0.5753	5.52e-09
10	5.95e-10	9.30e-13	-5.74e-03	-5.83e-03	4.067	0.3659	5.21e-09
11	5.45e-10	8.52e-13	-5.74e-03	-5.82e-03	4.104	0.0842	7.11e-09
12	2.82e-10	4.40e-13	-5.76e-03	-5.80e-03	4.399	0.4833	1.92e-09
13	3.57e-11	5.58e-14	-5.78e-03	-5.79e-03	4.982	0.8733	5.45e-11
14	2.36e-12	3.70e-15	-5.79e-03	-5.79e-03	5.968	0.9338	3.84e-12
15	4.01e-14	9.12e-17	-5.79e-03	-5.79e-03	7.076	0.9830	2.59e-14
16	9.57e-16	7.19e-17	-5.79e-03	-5.79e-03	8.717	0.9936	2.84e-16
17	6.96e-16	8.99e-17	-5.79e-03	-5.79e-03	11.016	0.9950	3.99e-18
18	5.33e-16	9.64e-17	-5.79e-03	-5.79e-03	13.313	0.9950	2.01e-20

optimization converged

counter	pri_inf	dual_inf	pri_obj	dual_obj	sigfig	alpha	nu

0	9.51e+01	1.05e-03	-1.14e+00	-5.01e+02	0.000	0.0000	1.00e+02
1	4.76e-01	5.26e-06	-1.14e+00	-9.08e+02	0.000	0.9950	2.07e-05
2	3.76e-03	4.16e-08	-1.04e+00	-2.00e+01	0.000	0.9921	5.16e-06
3	4.25e-04	4.70e-09	-1.07e+00	-3.22e+00	0.000	0.8870	2.06e-05
4	5.71e-05	6.32e-10	-1.12e+00	-1.43e+00	0.845	0.8656	3.29e-06
5	1.59e-05	1.76e-10	-1.14e+00	-1.22e+00	1.400	0.7217	2.05e-06
6	5.61e-07	6.21e-12	-1.14e+00	-1.15e+00	2.545	0.9647	8.83e-09
7	1.82e-07	2.01e-12	-1.14e+00	-1.14e+00	3.027	0.6767	5.60e-08
8	1.05e-07	1.16e-12	-1.14e+00	-1.14e+00	3.273	0.4240	6.15e-08
9	1.23e-08	1.36e-13	-1.14e+00	-1.14e+00	4.057	0.8821	1.34e-09
10	3.53e-09	3.90e-14	-1.14e+00	-1.14e+00	4.485	0.7139	1.34e-09
11	6.19e-10	6.86e-15	-1.14e+00	-1.14e+00	5.213	0.8244	1.85e-10
12	5.48e-11	6.00e-16	-1.14e+00	-1.14e+00	6.211	0.9115	8.64e-12
13	3.09e-13	1.93e-17	-1.14e+00	-1.14e+00	8.331	0.9944	3.47e-15
14	1.71e-15	2.36e-17	-1.14e+00	-1.14e+00	10.629	0.9950	2.07e-17
15	8.54e-16	2.47e-17	-1.14e+00	-1.14e+00	12.932	0.9950	1.04e-19

optimization converged

counter	pri_inf	dual_inf	pri_obj	dual_obj	sigfig	alpha	nu

0	9.51e+01	2.55e-02	-4.03e-02	-5.00e+02	0.000	0.0000	1.00e+02
1	6.65e-01	1.78e-04	-3.70e-02	-9.06e+02	0.000	0.9930	4.04e-05
2	3.32e-03	8.92e-07	7.37e-02	-1.96e+01	0.000	0.9950	2.13e-06
3	9.74e-05	2.61e-08	5.67e-02	-5.26e-01	0.259	0.9707	1.42e-06
4	3.25e-06	8.72e-10	-3.01e-02	-9.08e-02	1.229	0.9666	5.39e-08

5	6.83e-07	1.83e-10	-3.92e-02	-5.32e-02	1.870	0.7896	2.31e-07
6	2.92e-07	7.83e-11	-4.03e-02	-4.61e-02	2.254	0.5729	2.29e-07
7	1.66e-07	4.46e-11	-4.11e-02	-4.44e-02	2.497	0.4304	1.73e-07
8	5.72e-08	1.53e-11	-4.14e-02	-4.26e-02	2.951	0.6561	3.45e-08
9	1.22e-08	3.27e-12	-4.15e-02	-4.18e-02	3.614	0.7870	4.55e-09
10	9.66e-09	2.59e-12	-4.15e-02	-4.17e-02	3.716	0.2065	1.53e-08
11	4.87e-09	1.31e-12	-4.16e-02	-4.17e-02	4.017	0.4956	4.63e-09
12	1.12e-09	3.01e-13	-4.16e-02	-4.16e-02	4.577	0.7696	4.58e-10
13	2.14e-10	5.74e-14	-4.16e-02	-4.16e-02	5.232	0.8096	8.55e-11
14	2.71e-11	7.28e-15	-4.16e-02	-4.16e-02	6.072	0.8733	8.28e-12
15	3.49e-13	1.49e-16	-4.16e-02	-4.16e-02	7.768	0.9871	1.21e-14
16	3.38e-15	1.34e-16	-4.16e-02	-4.16e-02	9.190	0.9908	1.26e-16
17	8.25e-16	1.69e-16	-4.16e-02	-4.16e-02	10.323	0.9938	2.12e-18
18	6.06e-16	2.41e-16	-4.16e-02	-4.16e-02	12.205	0.9950	1.02e-19

optimization converged							
counter	pri_inf	dual_inf	pri_obj	dual_obj	sigfig	alpha	nu

0	9.51e+01	1.11e-02	-8.48e-02	-5.00e+02	0.000	0.0000	1.00e+02
1	6.46e-01	7.56e-05	-8.36e-02	-9.06e+02	0.000	0.9932	3.82e-05
2	3.23e-03	3.78e-07	-1.86e-02	-1.94e+01	0.000	0.9950	2.13e-06
3	7.94e-05	9.29e-09	-2.92e-02	-5.10e-01	0.330	0.9754	9.84e-07
4	5.01e-06	5.86e-10	-8.02e-02	-1.35e-01	1.295	0.9370	1.60e-07
5	1.46e-06	1.70e-10	-8.41e-02	-9.97e-02	1.841	0.7091	4.04e-07
6	1.33e-07	1.56e-11	-8.48e-02	-8.63e-02	2.846	0.9086	1.10e-08
7	1.59e-08	1.86e-12	-8.49e-02	-8.52e-02	3.657	0.8809	1.85e-09
8	7.86e-09	9.19e-13	-8.50e-02	-8.51e-02	3.926	0.5049	5.32e-09
9	3.48e-09	4.08e-13	-8.50e-02	-8.50e-02	4.279	0.5565	2.27e-09
10	3.12e-09	3.65e-13	-8.50e-02	-8.50e-02	4.314	0.1043	4.49e-09
11	7.64e-10	8.94e-14	-8.50e-02	-8.50e-02	4.881	0.7551	2.73e-10
12	1.52e-10	1.78e-14	-8.50e-02	-8.50e-02	5.569	0.8011	4.84e-11
13	7.32e-11	8.55e-15	-8.50e-02	-8.50e-02	5.891	0.5184	6.13e-11
14	5.68e-12	6.70e-16	-8.50e-02	-8.50e-02	6.704	0.9225	7.03e-13
15	1.11e-13	4.40e-17	-8.50e-02	-8.50e-02	7.953	0.9805	6.77e-15
16	1.68e-15	3.96e-17	-8.50e-02	-8.50e-02	9.457	0.9860	1.95e-16
17	7.24e-16	5.69e-17	-8.50e-02	-8.50e-02	11.743	0.9950	7.85e-19
18	6.16e-16	5.20e-17	-8.50e-02	-8.50e-02	14.050	0.9950	4.06e-21

optimization converged							

2.1.2 Calculate the gap for each state

Next, we will generate gaps for each donor state. Note: Here begins a divergence between Mixtape and my work. It is much easier to use a long format for the data instead of wide format.

2.2. CALCULATE PRE-TREATMENT AND POST-TREATMENT RMSPE41

We will generate a gap between treat and control for each year, and we will loop through each state and calculate a gap.

```
local statelist 1 2 4 5 6 8 9 10 11 12 13 15 16 17 18 20 21 22 23 24 25 26 27 28 ///
                29 30 31 32 33 34 35 36 37 38 39 40 41 42 45 46 47 48 49 51 53 55

foreach i of local statelist {
    use synth_bmprate_`i', clear
    keep _Y_treated _Y_synthetic _time
    drop if _time==.
    rename _time year
    rename _Y_treated treat
    rename _Y_synthetic counterfact
    gen gap=treat-counterfact
    sort year
    gen state = `i'
    save synth_gap_bmprate_`i', replace
}
```

Append all placebos to main treatment effect into a tempfile. Our placebo_pmprate.dta file will be in long format not wide format, which will save us a lot of headache below.

```
tempfile gap
save `gap', emptyok
clear

local statelist 1 2 4 5 6 8 9 10 11 12 13 15 16 17 18 20 21 22 23 24 25 26 27 28 ///
                29 30 31 32 33 34 35 36 37 38 39 40 41 42 45 46 47 48 49 51 53 55

foreach i of local statelist {
    append using synth_gap_bmprate_`i'
    save `gap', replace
}

save placebo_bmprate.dta, replace
```

2.2 Calculate pre-treatment and post-treatment *RMSPE*

We will calculate the exact p -values by estimating the pre-RMSPE and post-RMSPE. We will then calculate the ratio of the $\frac{\text{post-RMSPE}}{\text{pre-RMSPE}}$

```
sort state year
gen gap3 = gap*gap
```

```

by state: egen postmean=mean(gap3) if year>1993
by state: egen premean=mean(gap3) if year<=1993
gen rmspe=sqrt(premean) if year<=1993
replace rmspe=sqrt(postmean) if year>1993
gen ratio=rmspe/rmspe[_n-1] if 1994
gen rmspe_post=sqrt(postmean) if year>1993
gen rmspe_pre=rmspe[_n-1] if 1994

save synth_rmspe.dta, replace

keep if year == 1994

```

2.3 Sort and Rank

Next, we will sort and rank. We can use *gsort* for this task. We will get the rank by using $_n$ and total by using $_N$.

```

*Generate Rank and exact p-value
gsort -ratio
gen rank = _n
gen total = _N

```

2.4 Calculate exact p -values

Finally, we calculate the exact p -value for each state and plot the data.

```

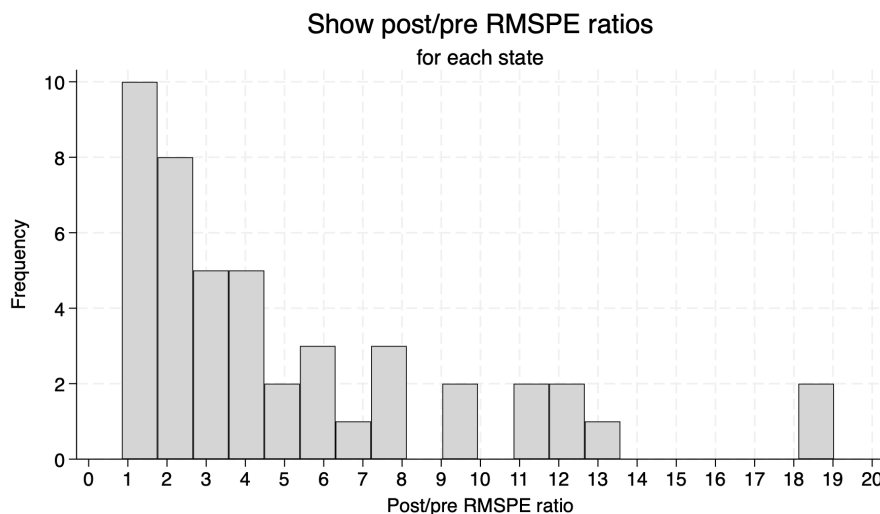
gen p=rank/total
histogram ratio, bin(20) frequency fcolor(gs13) lcolor(black) ylabel(0(2)10) ///
    xtitle(Post/pre RMSPE ratio) xlabel(0(1)20)
* Show the post/pre RMSPE ratio for all states, generate the histogram.
list rank p if state==48 // exact p-value for Texas is 0.04347

```

	rank	p

2.	2	.0434783

Our exact p -value for Texas is 0.04347, which is statistically significant at the 5% level.

Figure 2.1: Histogram of exact p -values

2.5 Plot all Placebos

Finally, we will plot the gaps in the outcomes for each state. I really don't feel like programming line gap year if state==n for each state So we will utilize macros Drop 1 and 55 from local

```
use placebo_bmprate.dta, clear

local statelist 1 2 4 5 6 8 9 10 11 12 13 15 16 17 18 20 21 22 23 24 25 26 27 28 ///
                29 30 31 32 33 34 35 36 37 38 39 40 41 42 45 46 47 48 49 51 53 55
```

We are going to loop through each state between 1 and 55 and append it

```
local graphappend " "
foreach s of local statelist {
    display "`s'"
    display "`graphappend'"
    *We want Texas (FIPS==48) to be a bit different, so we will use a if-else statement
    if `s' == 48 {
        local graph`s' " line gap year if state==`s',lp(solid) lw(thick) color(black) ||"
    }
    else {
        local graph`s' " line gap year if state==`s',lp(solid) lw(vthin) ||"
    }
}
```

```

local graphappend = "`graphappend'" + "`graph`s'"
}

local graphfinal = "`graphstate1'" + "`graphappend'" + "`graphstate55'" + ///
    ",legend(off) ytitle(Gap in Black Male Prisoner Rate) xtitle(Year)"
    + "xline(1993,lp(dash)) title(Gaps in Prisoner Rate)"
display "`graphfinal'"

twoway `graphfinal'

```

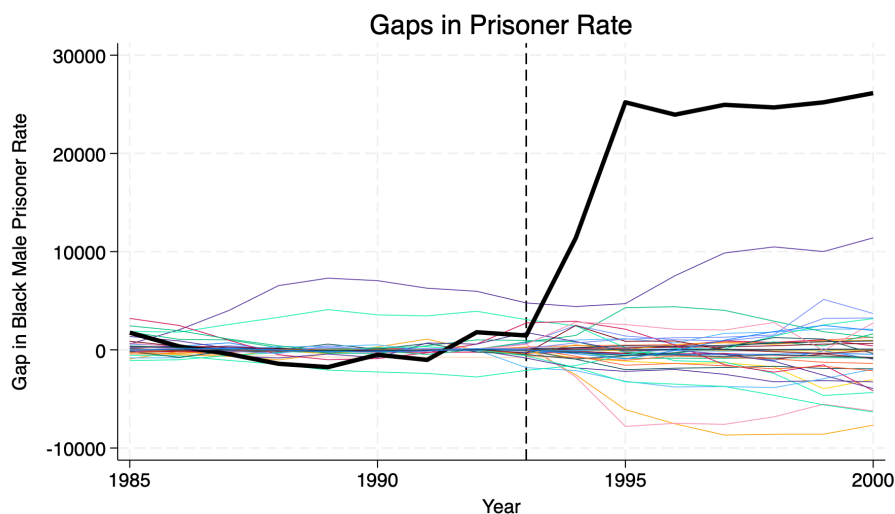


Figure 2.2: Gaps in Outcome for each state

2.6 Remove poor matches

Remember, Abadie, et al. (2010) suggest dropping any pre-treatment that is greater than 2 times your treatment state. Let's drop observations that were different from Texas in the pre-treatment period. We will drop the outliers *RMSPE* is 5 times more than Texas.

This means that we will drop:

- D.C. (11)
- Mississippi (28)
- Nevada (32)

- New Hampshire (33)
- Oregon (41)

We will grab our pre-treatment *RMSPE* that we got after we ran our placebo tests and compare them to Texas (42).

```
use synth_rmspe.dta, clear
keep if year==1993
keep state rmspe_pre

sort state
gen ratio_check = rmspe_pre/rmspe_pre[42] //Texas is the 42 observation
sort ratio_check
```

We see that DC and CA have 2 times the RMSPE in pre-treatment. An issue is that CA has important weight!

```
use placebo_bmprate.dta, clear

*I really don't feel like programming line gap year if state==n for each state
*So we will utilize macros
*Drop 1 and 55 from local
local statelist 1 2 4 5 8 9 10 12 13 15 16 17 18 20 21 22 23 24 25 26 27 28 ///
                29 30 31 32 33 34 35 36 37 38 39 40 41 42 45 46 47 48 49 51 53 55

*We are going to loop through each state between 1 and 55 and append it
local graphappend " "
foreach s of local statelist {
    display "`s'"
    display "`graphappend'"

    *We want Texas (FIPS==48) to be a bit different, so we will use a if-else statement
    if `s' == 48 {
        local graph`s' " line gap year if state==`s',lp(solid) lw(thick) color(black) ||"
        *tway line gap year if state==`s'
    }
    else {
        local graph`s' " line gap year if state==`s',lp(solid) lw(vthin) ||"
    }
    local graphappend = "`graphappend'" + "`graph`s'"
}

*Let's add some options - turn off legend and add titles
local graphfinal = "`graphstate1'" + "`graphappend'" + "`graphstate55'" + ///
                  ",legend(off) ytitle(Gap in Black Male Prisoner Rate) xtitle(Year)" ///
                  + "xline(1993,lp(dash)) title(Gaps in Prisoner Rate) subtitle(Without DC MS NV)"
```

```
twoway `graphfinal'
```

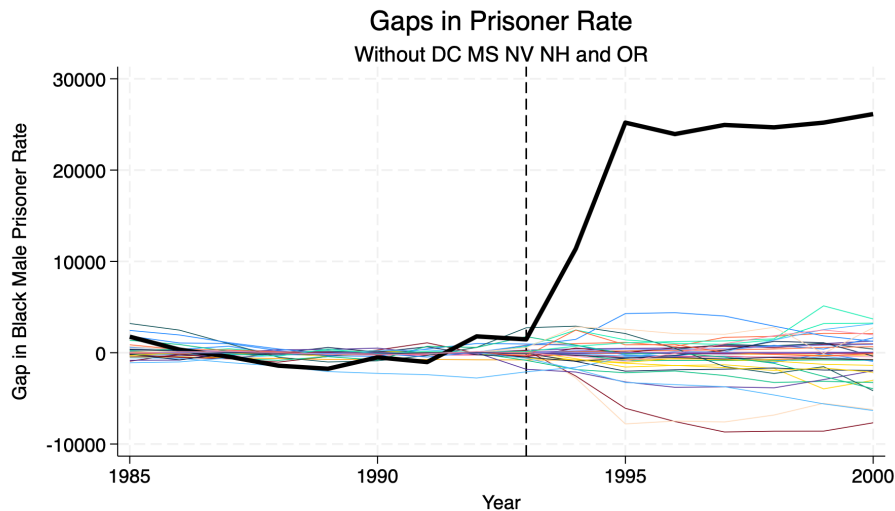


Figure 2.3: Gaps in Outcomes with DC, MS, NV, NH, and OR

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